



Decoding Social Intent from Neural Oscillations

Cogs 118C Final Project

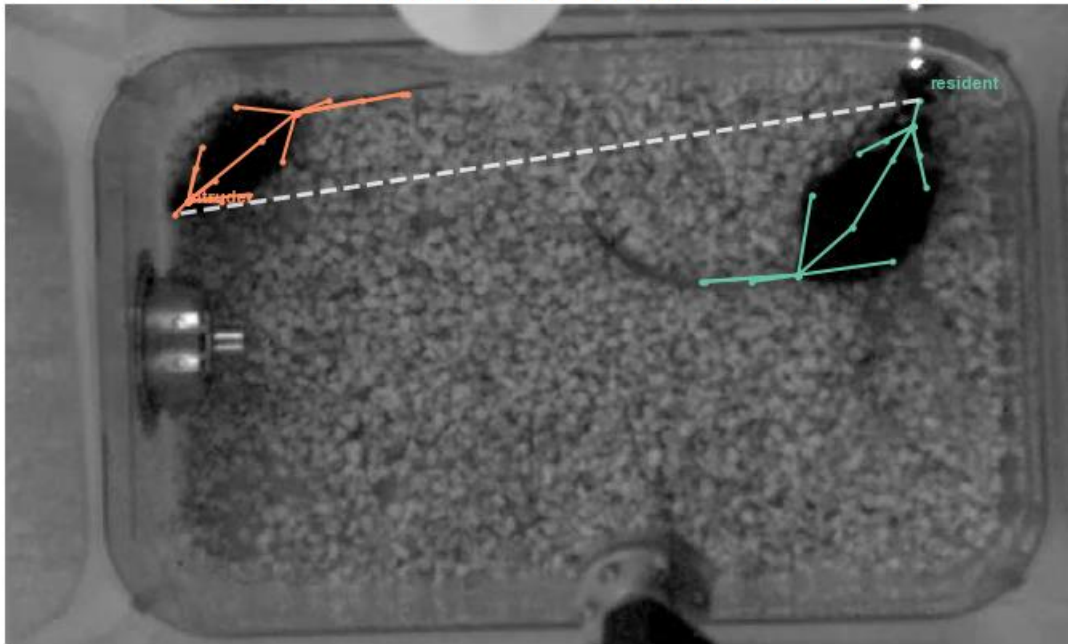
Kaiwen Bian; Ketan Mittal; Anirudh Annabathula; Yixuan Xin

Introduction

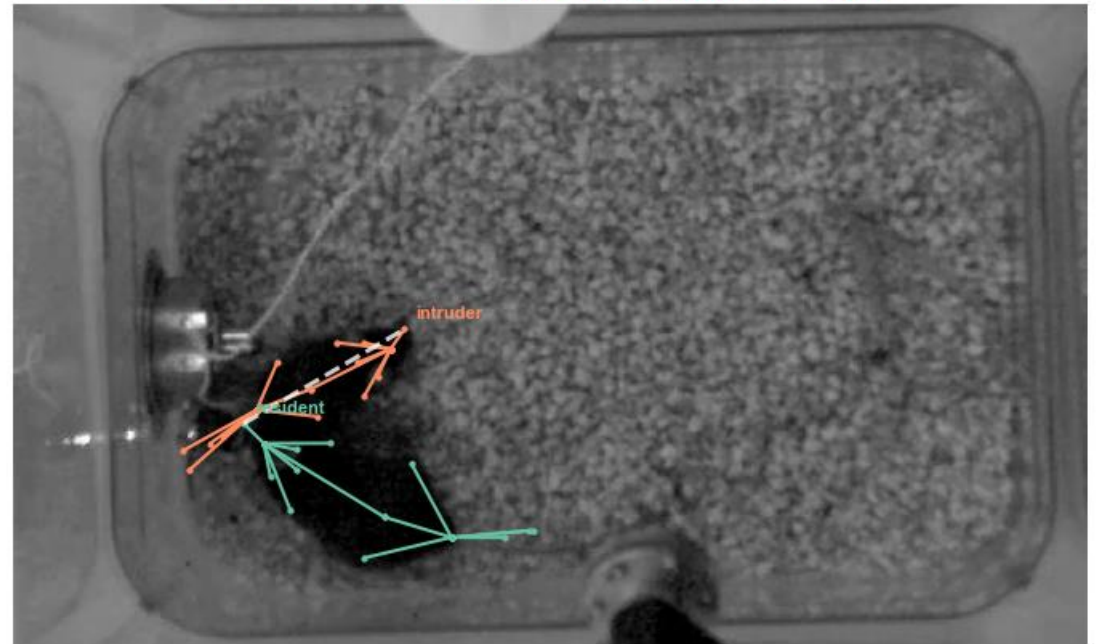
Decoding Social Intent from Neural Oscillations

What the social label means in the video

NOT SOCIAL — nose-to-body = 241px (threshold = 10px)



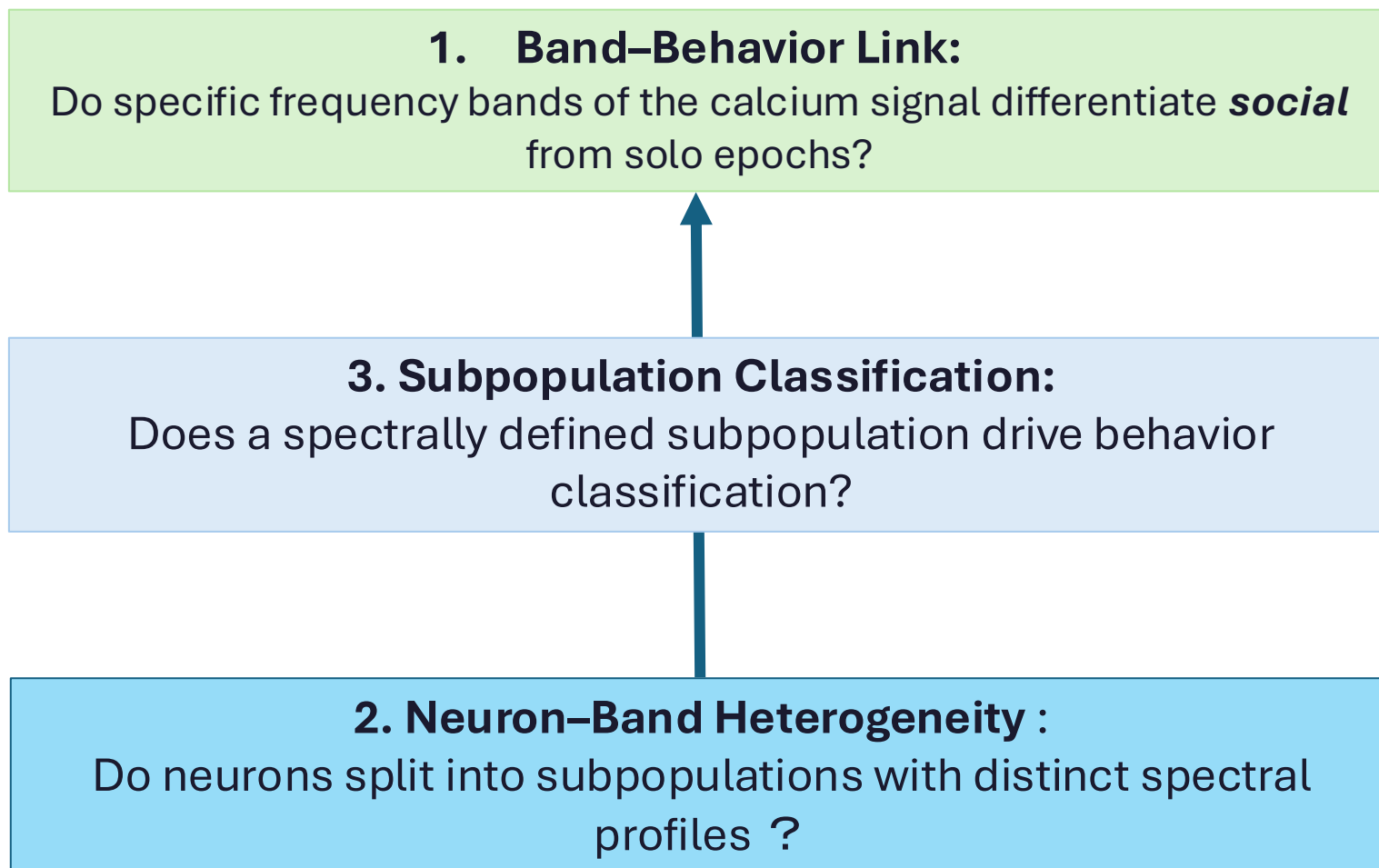
SOCIAL — nose-to-body = 9px (threshold = 10px)



Theta-band frequency brain waves in the prefrontal cortex have been linked to social behavior in mice ¹(LFP), and calcium imaging has revealed rich neural activity during social behavior using correlation studies ²

Workflows & Motivations

Decoding Social Intent from
Neural Oscillations



Dataset Overview

Decoding Social Intent from
Neural Oscillations

Feature	Description
● Calcium traces (dF/F)	Per-neuron fluorescence signal from prefrontal cortex
● Social bout labels	Binary social vs solo for each frame
● Isolation condition	GH (7d), GH (24hr), 24hr, or 7d per session
● Behavior video	Top-down arena recording at 25 fps
● Pose tracking (SLEAP)	Keypoint coordinates for subject + intruder
● Spatial footprints	Neuron ROI masks and locations in the FOV

18 sessions**3,938 neurons****86.7 min****30 Hz**

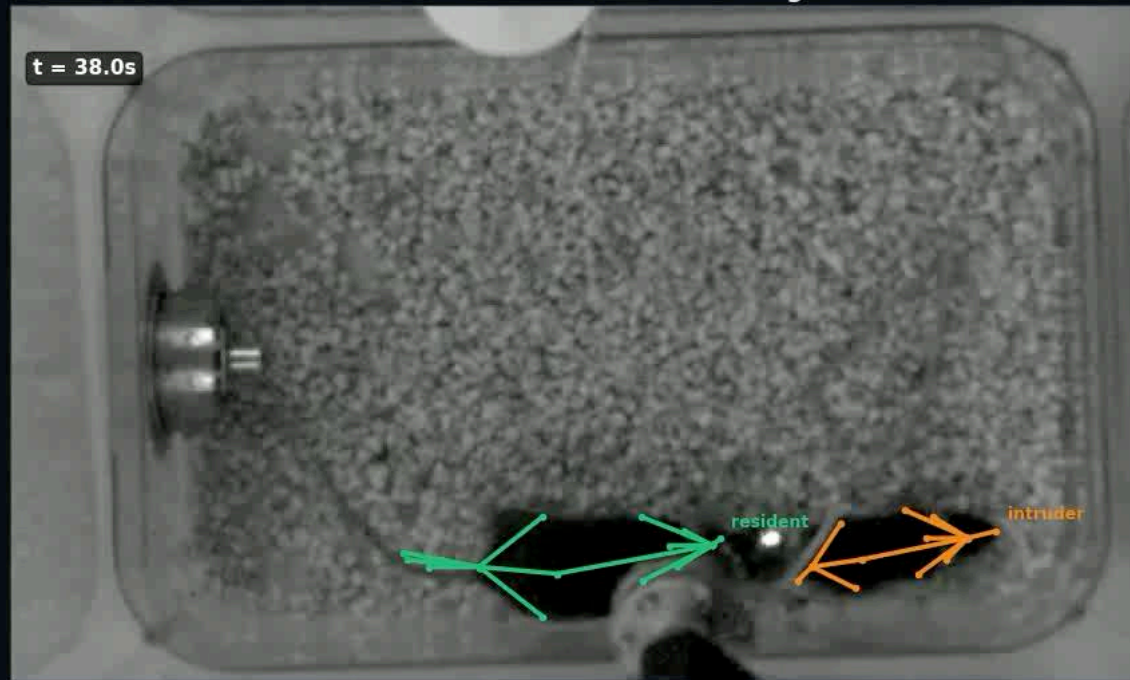


Frequency Bands and Band Power

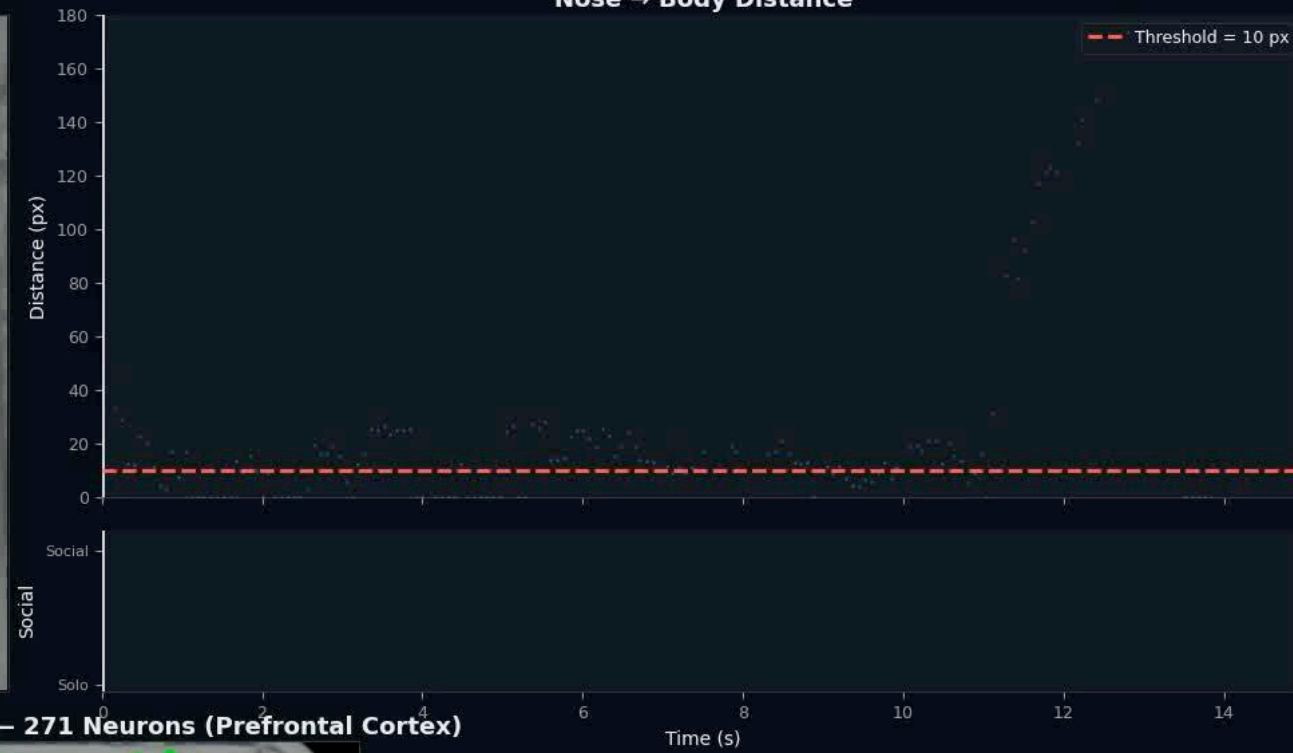
Band	Range	Period	Captures
Infraslow	0.01 – 0.1 Hz	10 – 100 s	Very slow drifts; partly photobleaching artifact
Slow	0.1 – 1.0 Hz	1 – 10 s	Network-level synchronization
Delta	1.0 – 4.0 Hz	0.25 – 1 s	Slow rhythmic activity
Theta	4.0 – 7.0 Hz	0.14 – 0.25 s	Social behavior and attention

Dataset Demo — Calcium Imaging + Behavior (Session 5)

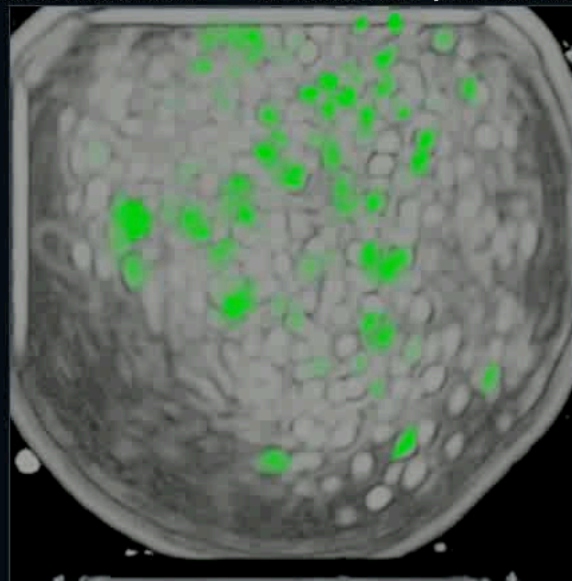
Behavior Video + SLEAP Tracking



Nose → Body Distance



Live Neural Activation — 271 Neurons (Prefrontal Cortex)



Methods & Results

Dissecting Signals

- FFT
- Butterworth band pass filtering + Hilbert Transform
- Band power (PSD via Welch's method)

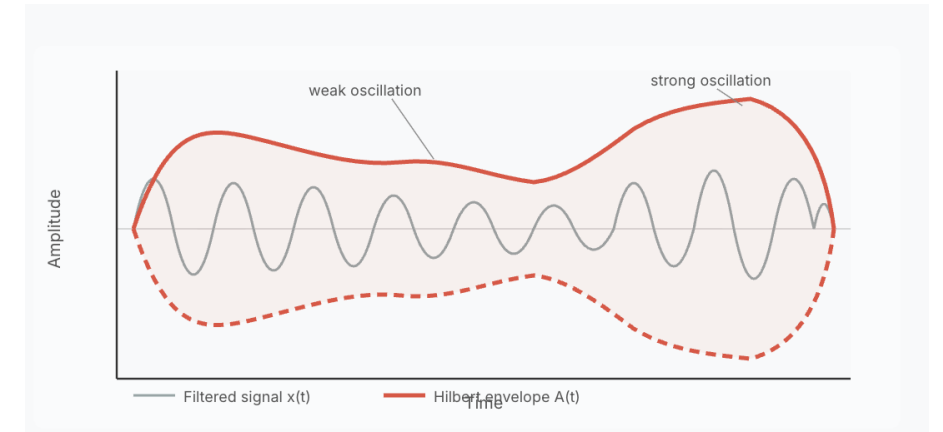
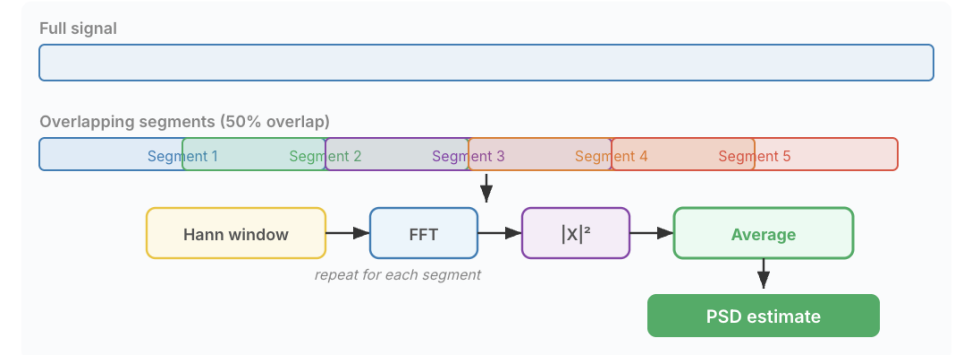
Using Signals

- K-means: frequency signal -> distinct neural population?
- Classifier: frequency signal -> social behavior?

Evaluations

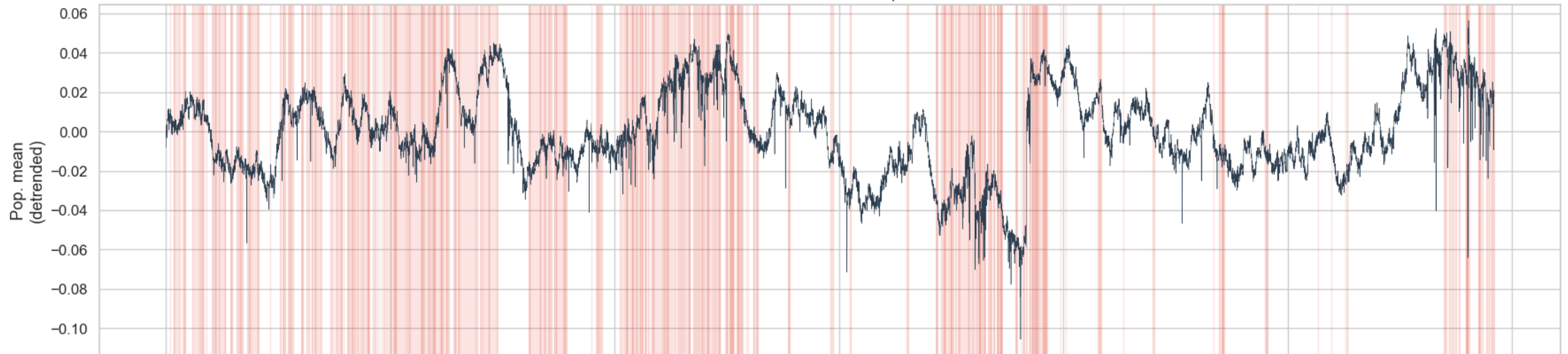
- Permutation Test
- Cohen's d Effect Size

Decoding Social Intent from Neural Oscillations



Dissecting Signals

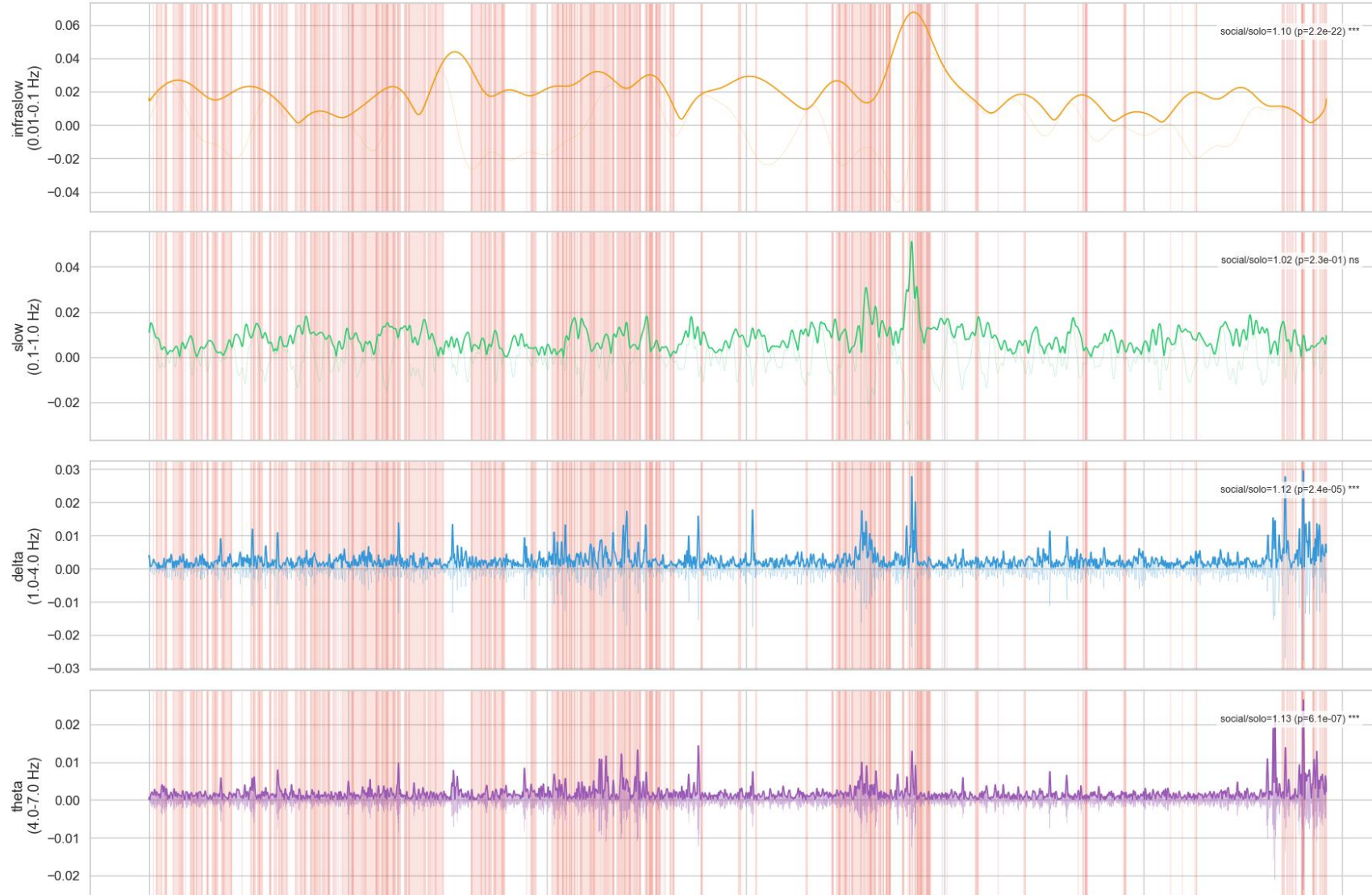
Decoding Social Intent from
Neural Oscillations



- A lot of rich information and seems to be periodic
- Can we dissect it? Can we aggregate among multiple neurons?

Dissecting Signals

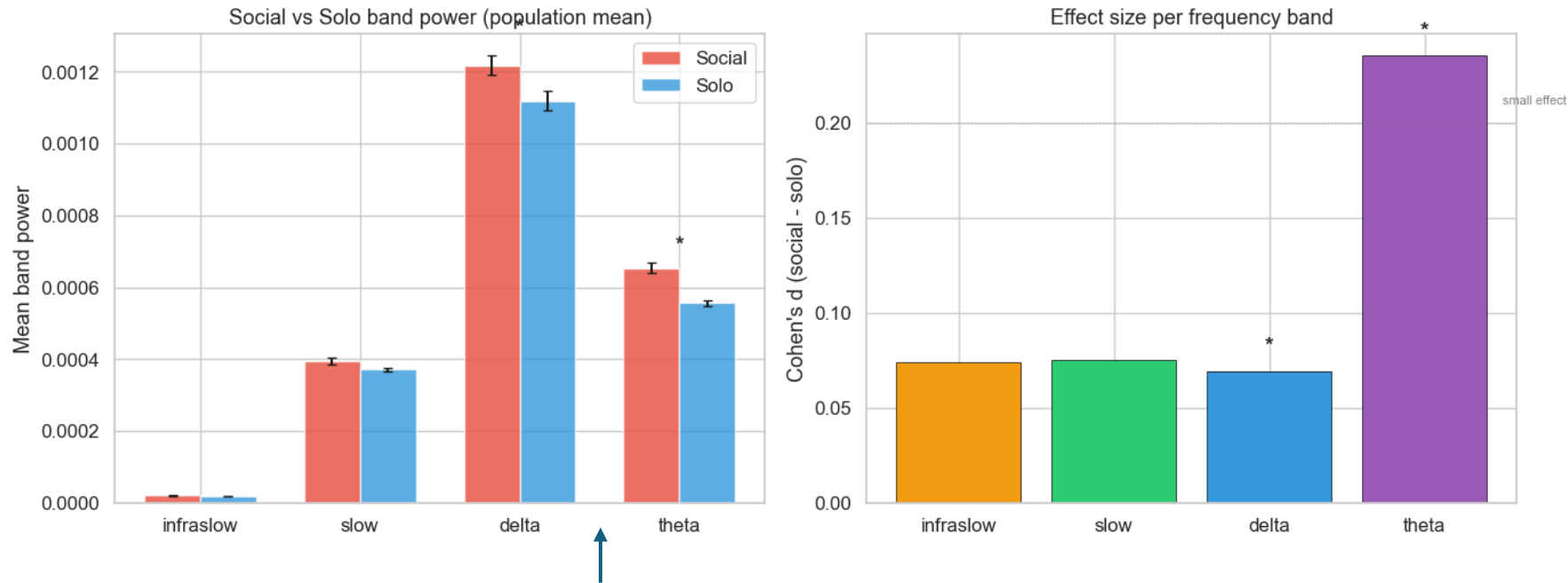
Decoding Social Intent from Neural Oscillations



Using Signals: Predicting Social Behaviors

Decoding Social Intent from Neural Oscillations

Welch PSD estimates for every neuron in every 1-second epoch, then compared band power between social and solo windows.

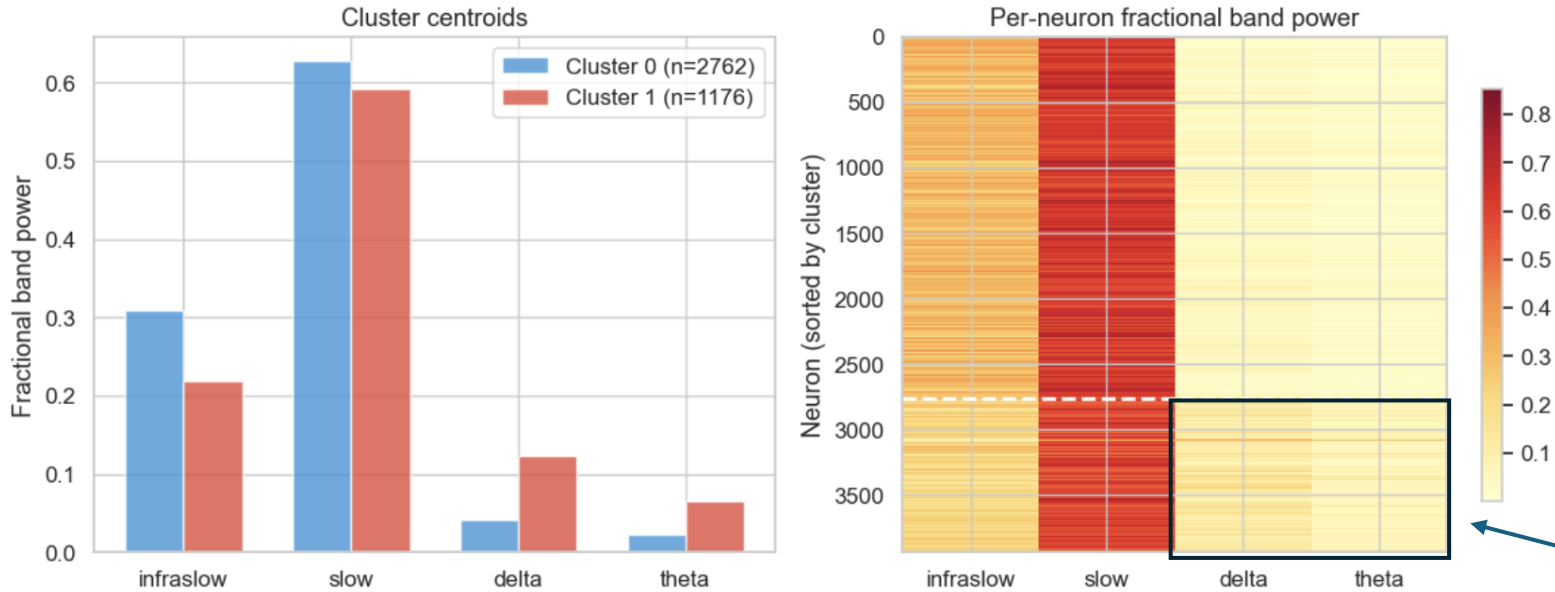


Seems to have greater difference (significant)

Theta-band power is systematically higher during social interaction, consistent across 14 of 18 sessions. This aligns with electrophysiology findings (Kuga et al. 2022), confirming theta signatures are recoverable from calcium imaging despite its limited bandwidth.

Using Signals: Cluster of Neural Population

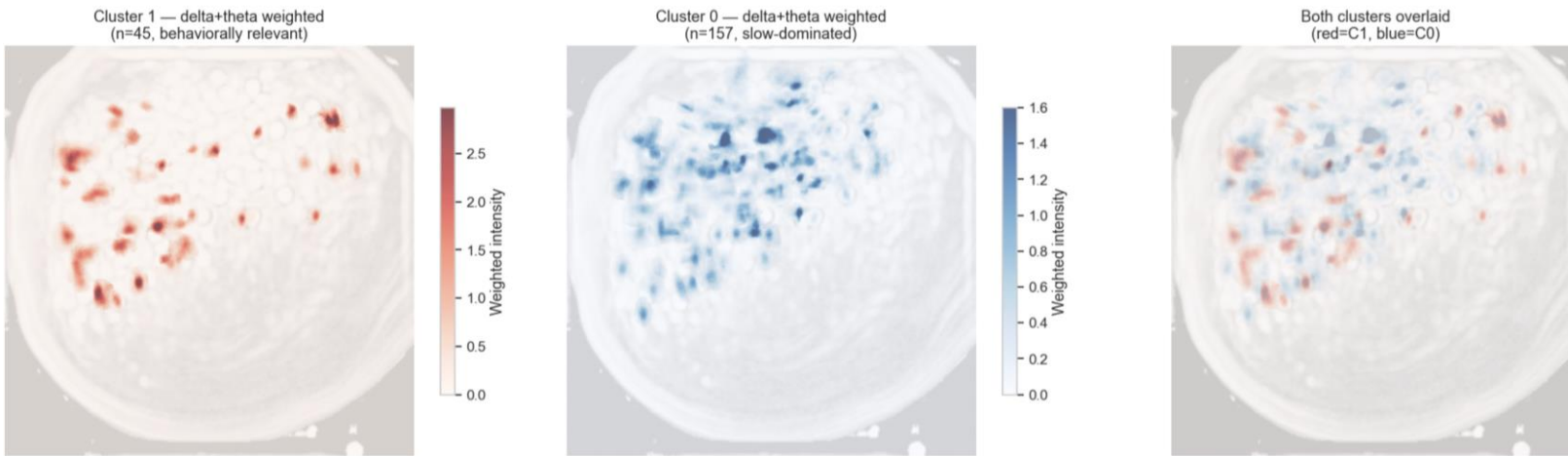
Decoding Social Intent from
Neural Oscillations



Cluster 0 (70%) is dominated by infra-slow and slow power.

Cluster 1 (30%) is dominated by delta and theta power.

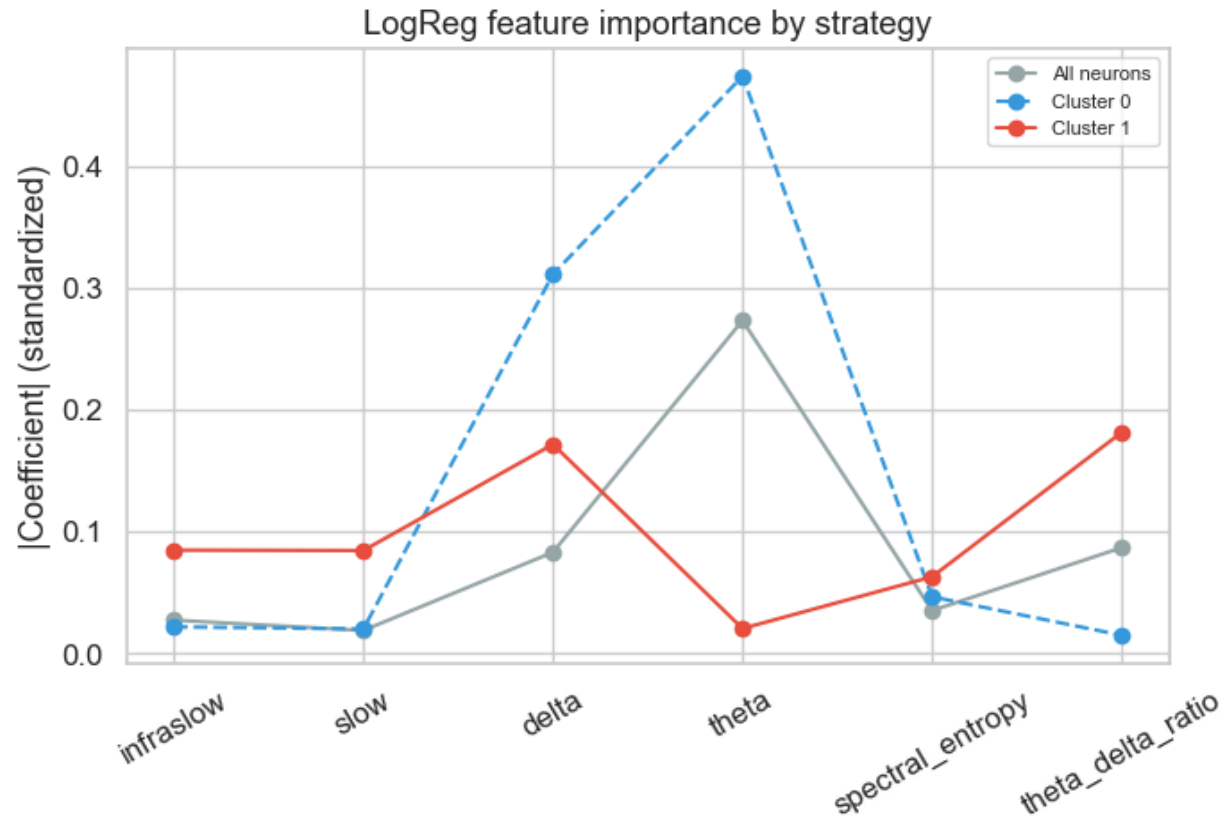
Only cluster two has
delta / theta power



Using Signals: Cluster of Neural Population

Decoding Social Intent from
Neural Oscillations

We trained linear classifiers (LDA, SVM, Logistic Regression) on spectral features extracted from each cluster separately



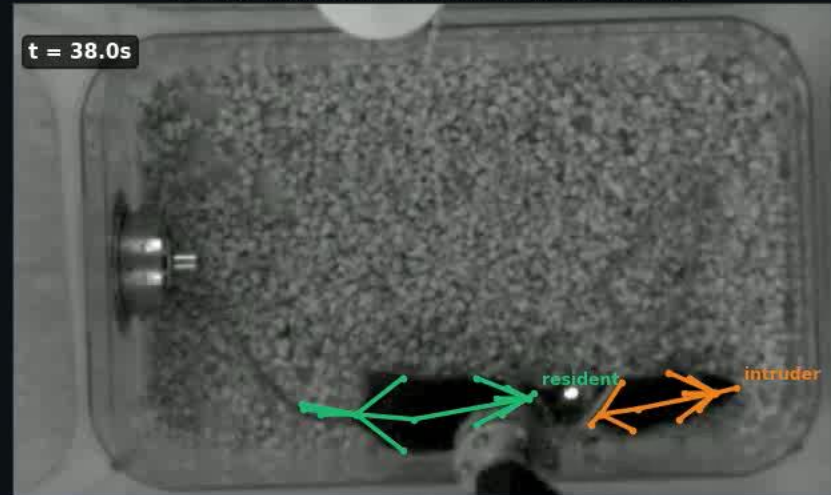
$$\mathbf{x} = \begin{bmatrix} P_{\text{infraslow}} \\ P_{\text{slow}} \\ P_{\text{delta}} \\ P_{\text{theta}} \\ H_{\text{spectral}} \\ P_{\text{theta}}/P_{\text{delta}} \end{bmatrix}$$

Somewhat counter intuitive:

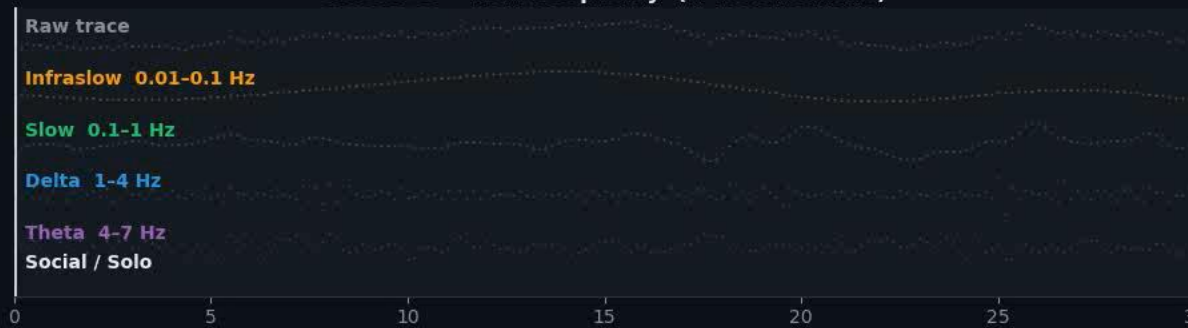
Cluster 1 uses theta/delta ratio (spectral shape), while the full population or Cluster 0 relies on raw theta power.

Neural Activity During Social Interaction

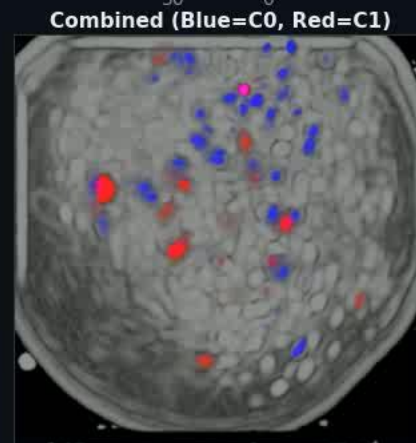
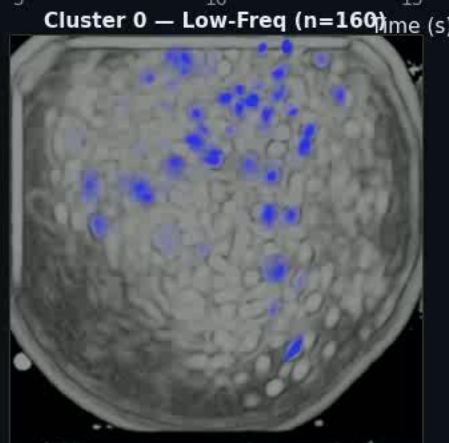
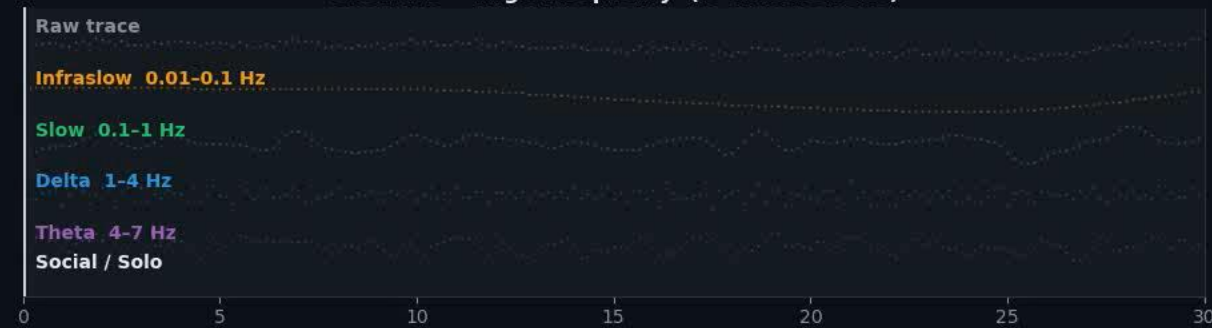
SLEAP Tracked Behavior (Session 5)



Cluster 0 — Low-Frequency (n=160 neurons)



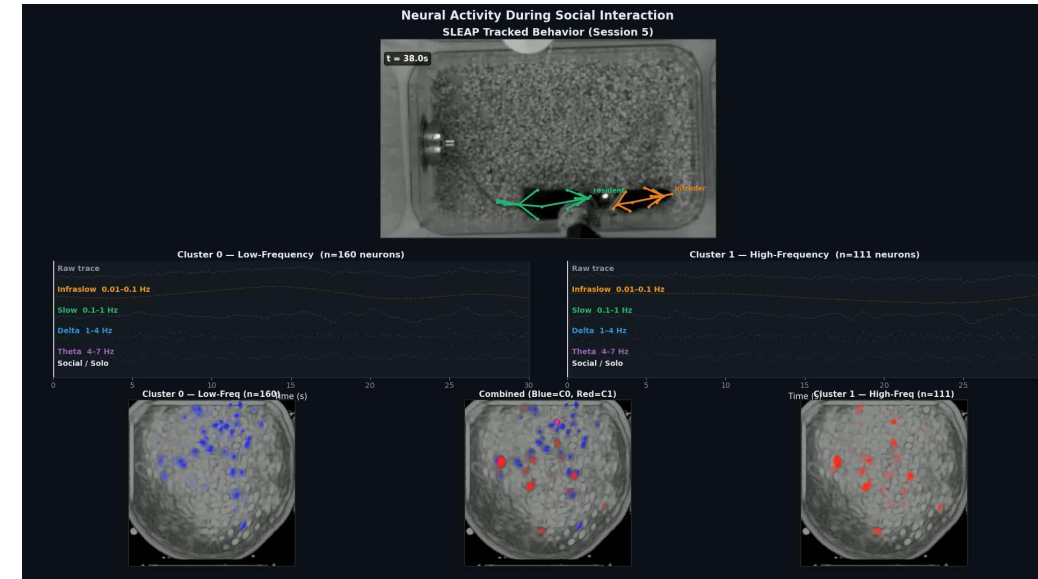
Cluster 1 — High-Frequency (n=111 neurons)



Discussions

Decoding Social Intent from Neural Oscillations

1. **Theta-band power** increases during social interaction.
2. This signal is not uniformly distributed across neurons, **only a minority** subpopulation (30%) is related to slow waves.
3. Slow waves are sufficient for above-chance **behavioral classification**...
4. Yet, the minority neurons associated with slow waves cannot produce reliable behavior classifications using slow waves alone.



Limitations

Current Limitations

Limits of Calcium
Imaging

Single Cortical Region
(prefrontal cortex)

Social Behavior with
singular Intruder mouse



Future Steps

Electrophysiology and
Calcium Imaging

Multiple Region Imaging
(prefrontal + amygdala)

Social Behavior with
colony/multiple mice

1. Frost NA, Haggart A, Sohal VS (2021). Dynamic patterns of correlated activity in the prefrontal cortex encode information about social behavior. *PLOS Biology*, 19(5), e3001235.
2. Kingsbury L, Huang S, Wang J, et al. (2019). Correlated neural activity and encoding of behavior across brains of socially interacting animals. *Cell*, 178(2), 429–446.e16.
3. Kuga N, Abe R, Takano K, Ikegaya Y, Sasaki T (2022). Prefrontal-amygdalar oscillations related to social behavior in mice. *eLife*, 11, e78428.
4. Liang B, Zhang L, Barbera G, et al. (2018). Distinct and dynamic ON and OFF neural ensembles in the prefrontal cortex code social exploration. *Neuron*, 100(3), 700–714.e9.
5. Mohapatra AN, Jabarin R, Ray N, Netser S, Wagner S (2025). Impaired emotion recognition in *Cntnap2*-deficient mice is associated with hyper-synchronous prefrontal cortex neuronal activity. *Molecular Psychiatry*, 30, 1440–1452.
6. Pereira TD, et al. EDGE: Education in Data and Guided Exploration. Salk Institute / UC San Diego. Calcium imaging and behavioral tracking datasets used in this project.
7. Tibau E, Valencia M, Soriano J (2013). Identification of neuronal network properties from the spectral analysis of calcium imaging signals in neuronal cultures. *Frontiers in Neural Circuits*, 7, 199.
8. Zhu L, Lee CR, Margolis DJ, Najafizadeh L (2018). Decoding cortical brain states from widefield calcium imaging data using visibility graph. *Biomedical Optics Express*, 9(7), 3017–3036.

Thank you!

Any Questions?

Contact us!

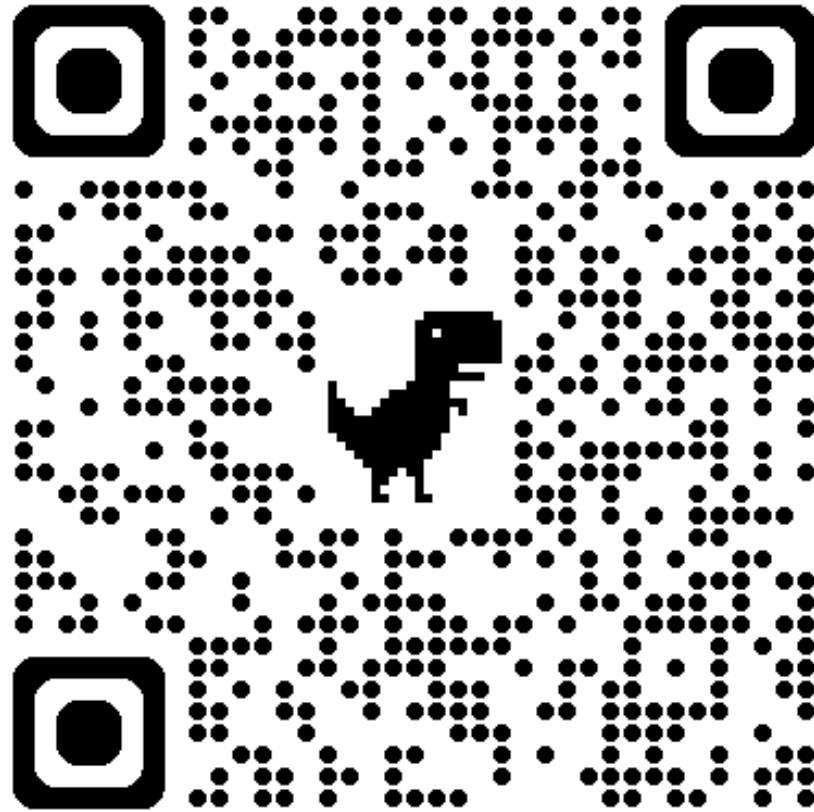
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https://kbian.org/neural_signal/



Introduction/Motivation (Ivy's Slides):

Hello everyone! As we can see, the video in the background shows a sample of our data. So this data is from the Talmo lab in the Salk Institute, and they use SLEAP which is a deep learning tool to track pose. In this experiment, a resident mouse is first alone in an arena, and later an intruder mouse is introduced. We label behavior as social or non-social. So social is defined when the resident mouse's nose is close to the intruder's body.

We are interested in decoding social intent from neural oscillation signals.

So we have previous studies shown that theta-band oscillations in the prefrontal cortex are linked to social behavior in mice. In the sample, calcium imaging helps us to record activity from many individual neurons, showing a rich neural dynamics in social interaction.

However, most of them analyze firing rates or correlations, instead of the frequency structure of the signals.

So in this project, we are interested in, first do specific frequency bands of the calcium signal distinguish social from non-social behavior? Second, do neurons are divided into subpopulations like on and off, high and low in distinct spectral profiles? Together, can these spectral subpopulations help to classify social behavior?

Dataset Overview (Ani's Slides):

We used a calcium and imaging behavioral dataset from Talmo's Lab at Salk. Here are the main components of our dataset. Calcium traces are the signals from individual neurons in the prefrontal cortex. They're our main neural measure. Social bout labels: frame labels for social vs alone. Isolation condition: how long each mouse was alone before socialization. Behavior video: shows the mice which is used with pose tracking with SLEAP, a tool to track the movements of the mice precisely. Spatial footprints: we have ROI masks for each neuron and information on where the neuron is located. Here are the four bands that we used in our analysis. These bands are differentiated in the speed of their oscillations. The most relevant one for our purposes was the theta band, as it has been shown in literature that that theta oscillations in the prefrontal cortex track social behavior and attention in mice. Putting all of our data together, the top left is a video with SLEAP tracking keypoints overlayed on the mice in the frame. On the right, we have a graph of the body distance between the mice. More specifically, if the nose of the intruder mouse was less than the 10 pixels of the other mouse, this was a previously established threshold, it would be marked as a social period. The bottom image shows calcium imaging field from the same session as above, where the green regions of interests are the activity of the neurons.

Methods and Results (Ketan's Slides):

Let's dive into the methods and results. To begin the analysis, we first decompose the neural signal using a Fast Fourier Transform – or FFT – which allows us to sort neural activity into specific frequency bands. From there, we applied Butterworth band-pass filtering and the Hilbert transform to isolate and visualize the signals, separate signal from noise, and remove baseline drift. In simpler terms, the Hilbert transform converts a bandpass-filtered signal into a smooth envelope that gives us a frame-by-frame measure of band-specific neural activity. We also computed band power using Power Spectral Density – PSD -- estimated via Welch's method. On the right, you can see a figure illustrating this pipeline, using one-second time windows to segment the data. With these processed signals, we can identify distinct neural populations by applying k-means clustering to separate neurons into two groups. We then trained a classifier to distinguish social behaviors based solely on their frequency-domain signatures. To evaluate our approach, we used permutation testing and Cohen's d as our statistical metrics.

This figure shows the population-level mean neural signal. The red segments mark periods of social behavior, and you can see the signal is highly periodic during these windows – suggesting there are frequency components that may reliably correlate with social activity. We can also break this down further to identify which individual neurons are tuned to which frequencies, giving us a clearer picture of how neural populations encode social information.

Here we show how we decomposed the signal into its constituent frequency bands. As Ani mentioned earlier, we work with four bands spanning 0.01 Hz to 7 Hz — 7 Hz being the maximum resolvable frequency given calcium imaging's sampling rate. This upper bound is important: calcium imaging has a relatively slow sampling rate and a biological limit, so any neural activity above 7 Hz would be aliased and appear as artifactual low-frequency content, which is why we strictly restrict our analysis to this range. As you move to higher frequencies within these bands, the signal becomes more fine-grained, and we start to see hints of a relationship between frequency content and social behavior. While the pattern may be subtle visually, over the next few slides we'll walk through the numerical evidence showing that certain frequency bands reliably predict social behavior.

Methods Continued (Kevin's Slides):

Remaining the audience of the three questions we have.

- Answer question 1 -> there are significant difference in the social and solo delta/theta band distribution with theta band having the greatest effect size
- Answering question 2 -> different clusters of neurons do fire differently, preferring different bands of frequency
 - Explain the RHS figure first
 - Explain that blue is higher than red for infraslow/slow and red is higher than blue for delta/theta
 - The clustering image will be shown in more details later

- Answering the third question -> we did not find direct evidence that cluster 1 (correlating more to fast theta wave) as the most effective feature in predicting behaviors, the A -> B -> C logic is not 100% established.
 - Explain what the features are

Discussion (Kevin's Slides):

Just rementioning the three points, summarizing our three answers

Limitations (Ketan's Slides):

A few limitations worth noting. First, calcium imaging itself imposes a hard frequency ceiling of 7 Hz, which means we cannot resolve higher-frequency oscillations – such as gamma band activity, which we discussed earlier in class. Second, our current model only looks at a single cortical region: the prefrontal cortex. Additionally, our data only shows how a singular mouse behaves when they encounter an intruder mouse in their territory. For future work, the addition of multiple scenarios – such as introducing a mouse to a colony, introducing a colony to a mouse, and introducing mice of different sexes to each other – will allow for further analysis and demonstrate greater neural understanding in the final results. The most immediate next step is expanding to multiple brain regions – incorporating areas like the prefrontal cortex alongside the amygdala – to strengthen our results and better pinpoint which specific neural populations drive social behavior. We also want to explore combining electrophysiology with calcium imaging, which would give us much higher temporal resolution and allow us to capture the richer frequency dynamics that calcium imaging alone can't resolve.

Ending (Ketan's Slides):

Thank you! Here are our references. We're happy to take any questions. And finally, if you want to explore our project in more depth – with additional figures, methods details, and results – we have a website linked here, along with a QR code you can scan to access it directly.

Decoding Social Intent from Neural Oscillations

Spectral analysis of calcium imaging reveals a theta-enriched neuron subpopulation that encodes social behavior

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COGS 118C · Winter 2026

Website: https://kbian.org/neural_signal/index.html

Abstract

Social behavior requires coordinated neural activity, yet whether spectral signatures of individual neurons can predict social state remains unclear. We analyzed calcium imaging recordings from mouse prefrontal cortex across 18 sessions, extracting frequency-band power features from 3,938 neurons. While theta-band (4–7 Hz) power increases during social interaction at the population level (Cohen's $d = 0.235$, $p < 0.001$), classification using all neurons fails a permutation test. Spectral clustering reveals two subpopulations: a majority (70%) dominated by infraslow oscillations, and a minority (30%) enriched in delta and theta power. Only this minority subpopulation significantly classifies social vs. solo behavior (AUC = 0.570, permutation $p = 0.020$), using theta/delta ratio—not raw power—as its top feature. These results suggest that social coding is concentrated in a spectrally distinct neuronal minority.

Introduction

Theta-band (4–7 Hz) oscillations in the prefrontal cortex track social behavior in mice. Kuga et al. (2022) showed this with electrophysiology and confirmed it causally via optogenetics, while Mohapatra et al. (2025) demonstrated that excess theta synchrony impairs social discrimination. But these findings come from local field potentials—a macroscopic signal. We asked: **can the same theta signatures be recovered from individual neurons using calcium imaging?**

Calcium imaging studies of social behavior in mPFC have revealed rich single-cell coding (Liang et al. 2018; Kingsbury et al. 2019; Frost, Haggart & Sohal 2021), but every study uses rate-based or correlation-based measures—none have examined frequency content. The technical challenge is real: GCaMP6s has a ~600 ms decay time, placing theta at the very edge of what calcium imaging can resolve.

We bridge this gap using data from the EDGE neuroscience course (Talmo Pereira and colleagues), applying spectral feature extraction—Welch PSD, band power, theta/delta ratio—to single-neuron calcium traces. Rather than asking *which neurons fire more*, we ask *which neurons oscillate differently*.

Research Questions

1. **Band–Behavior Link** — Do specific frequency bands of the calcium signal differentiate social from solo epochs?
 2. **Neuron–Band Heterogeneity** — Do neurons split into subpopulations with distinct spectral profiles?
 3. **Subpopulation Classification** — Does a spectrally defined subpopulation drive behavior classification?
-

1. Setup & Data Loading

```
In [1]: import sys
sys.path.insert(0, '..')

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import h5py
from pathlib import Path

from scipy.signal import welch, detrend, butter, sosfiltfilt, hilbert
from scipy.stats import ranksums, mannwhitneyu, entropy
from scipy.ndimage import center_of_mass

from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis
from sklearn.svm import SVC
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import GroupKFold
```

```

from sklearn.metrics import roc_auc_score, accuracy_score

from tqdm import tqdm

# Project modules
from src.constants import (BEHAVIOR_FPS, IMAGING_FPS, FREQ_BANDS, BAND
                           ISO_COLORS, MIN_BOUT_S, PRE_S, POST_S)
from src.data import (download_data, load_entrances, load_behavior,
                      load_imaging, align_all_sessions, get_epoch_dura
                      load_spatial_footprints)
from src.signal_processing import (bandpass_filter, compute_welch_psd,
                                   compute_band_powers, band_envelope)
from src.analysis import (extract_labeled_windows, bp_list_to_array, c
                           compute_neuron_spectral_profiles, cluster_ne
                           extract_feature_matrix, run_classification,
                           run_permutation_test)
from src.visualization import shade_social, add_band_shading

sns.set_theme(style='whitegrid', context='notebook', font_scale=1.1)
plt.rcParams['figure.figsize'] = (14, 5)
plt.rcParams['figure.dpi'] = 100
np.random.seed(42)

BAND_NAMES = list(FREQ_BANDS.keys())
iso_colors = ISO_COLORS
print('All imports successful.')

```

All imports successful.

```

In [2]: # Download data from Google Drive (skips if already present)
DATA_DIR = download_data()

# Load session metadata
entrances = load_entrances(DATA_DIR)
n_sessions = len(entrances)
print(f'Sessions: {n_sessions}')
print(f'Isolation conditions: {entrances["Isolation Length"].value_cou

[skip] calcium.00.h5 (238.6 MB)
[skip] social_bouts.00.h5 (0.5 MB)
[skip] SI3_2022_Entrance_Frames.xlsx (0.0 MB)
[skip] behavior_video.mp4 (37.6 MB)
[skip] behavior_tracking.slp (10.1 MB)
Sessions: 18
Isolation conditions: {'24hr': 6, '7d': 6, 'GH (7d)': 3, 'GH (24hr)':
3}

```

```

In [3]: # Load behavioral labels and calcium imaging
behavior = load_behavior(n_sessions, DATA_DIR)
imaging = load_imaging(n_sessions, DATA_DIR)

# Align behavior (25 fps) to calcium imaging (30 fps)
aligned_calcium, aligned_behavior, session_info, session_df = \

```

```
align_all_sessions(imaging, behavior, entrances)
```

```
Session 0: 7788 frames, 202 neurons, 260s, 38.9% social, GH (7d)
Session 1: 8776 frames, 12 neurons, 293s, 41.5% social, GH (7d)
Session 2: 8847 frames, 289 neurons, 295s, 16.3% social, GH (7d)
Session 3: 8887 frames, 268 neurons, 296s, 25.3% social, 24hr
Session 4: 8895 frames, 134 neurons, 296s, 30.4% social, 24hr
Session 5: 8882 frames, 271 neurons, 296s, 38.0% social, 24hr
Session 6: 8897 frames, 218 neurons, 297s, 21.6% social, 7d
Session 7: 8872 frames, 207 neurons, 296s, 30.7% social, 7d
Session 8: 8830 frames, 332 neurons, 294s, 32.5% social, 7d
Session 9: 8882 frames, 396 neurons, 296s, 29.7% social, GH (24hr)
Session 10: 8860 frames, 288 neurons, 295s, 24.1% social, GH (24hr)
Session 11: 6544 frames, 96 neurons, 218s, 36.2% social, GH (24hr)
Session 12: 8847 frames, 72 neurons, 295s, 25.1% social, 24hr
Session 13: 8904 frames, 225 neurons, 297s, 28.0% social, 24hr
Session 14: 8874 frames, 231 neurons, 296s, 30.5% social, 24hr
Session 15: 8798 frames, 235 neurons, 293s, 27.0% social, 7d
Session 16: 8835 frames, 176 neurons, 294s, 27.3% social, 7d
Session 17: 8832 frames, 286 neurons, 294s, 28.7% social, 7d
```

Aligned 18 / 18 sessions.

```
In [4]: # Dataset summary
total_neurons = sum(c.shape[1] for c in aligned_calcium)
total_frames = sum(c.shape[0] for c in aligned_calcium)
total_minutes = total_frames / IMAGING_FPS / 60

print(f'Total neurons: {total_neurons:,}')
print(f'Sessions used: {len(aligned_calcium)}')
print(f'Total recording time: {total_minutes:.1f} min')
print(f'Sampling rate: {IMAGING_FPS} Hz')
```

```
Total neurons: 3,938
Sessions used: 18
Total recording time: 86.7 min
Sampling rate: 30 Hz
```

```
In [5]: # Pick example session closest to 30% social
ex = session_df['social_frac'].sub(0.3).abs().idxmin()
ex_cal = aligned_calcium[ex]
ex_beh = aligned_behavior[ex]
ex_info = session_info[ex]
t = np.arange(len(ex_cal)) / IMAGING_FPS

print(f'Example: session {ex_info["session_idx"]} | '
      f'{ex_info["n_neurons"]} neurons | {ex_info["duration_s"]:.0f}s'
      f'{ex_info["social_frac"]*100:.1f}% social | {ex_info["isolation"]:.1f}% isolation')
```

```
Example: session 9 | 396 neurons | 296s | 29.7% social | GH (24hr)
```

2. The Neural Recording

While the mouse behaves, a miniature microscope images calcium fluorescence through a GRIN lens implanted in the prefrontal cortex. Neurons that fire produce transient increases in fluorescence ($\Delta F/F$), giving us a continuous signal for each cell at 30 fps.

Figure 4 — Raw Calcium Traces

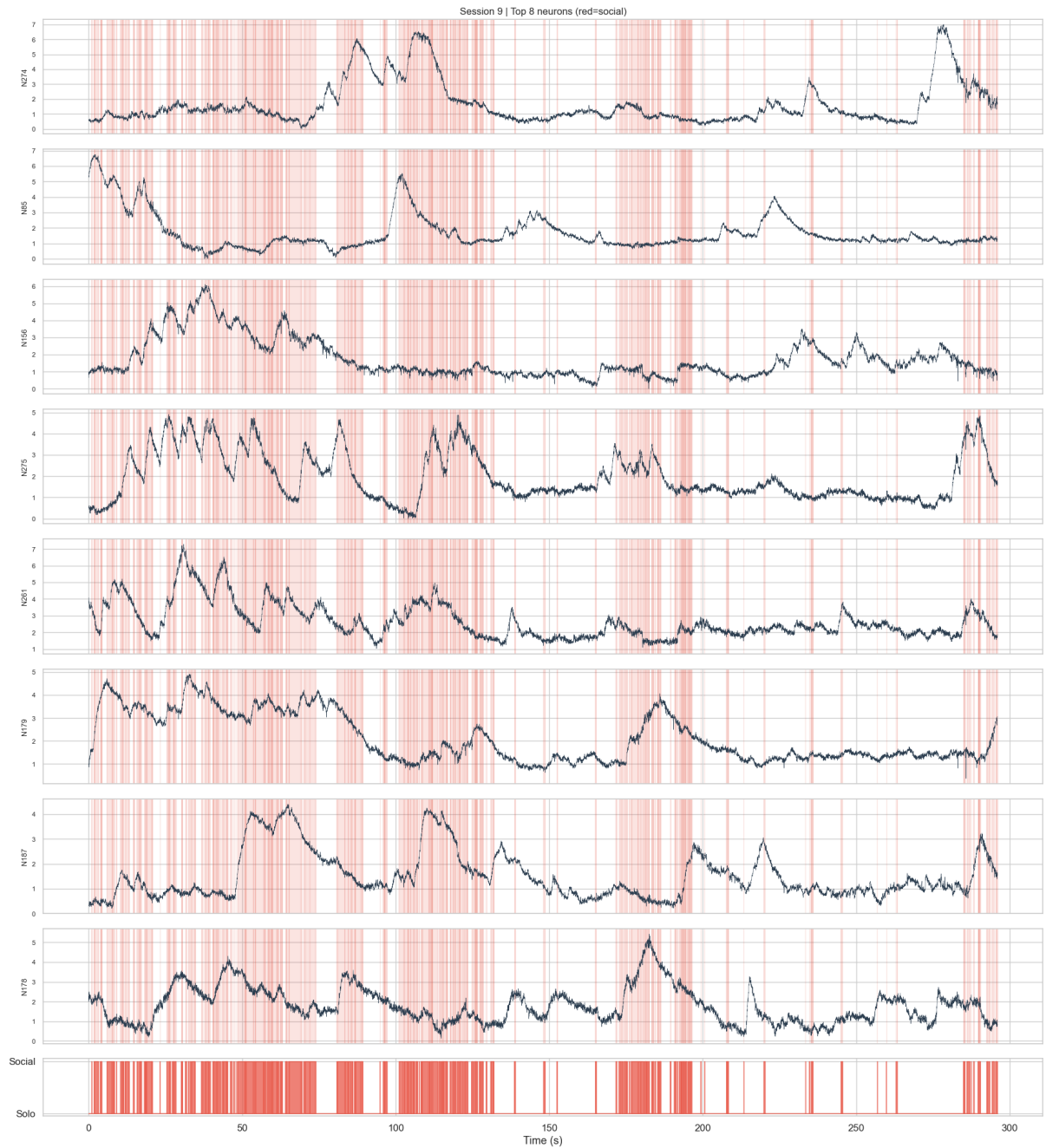
```
In [6]: # Figure 4: Top active neuron traces with social epoch shading
n_show = 8
top_neurons = np.argsort(ex_cal.std(axis=0))[-n_show:][::-1]

fig, axes = plt.subplots(n_show+1, 1, figsize=(18, 2.2*(n_show+1)),
                        sharex=True,
                        gridspec_kw={'height_ratios': [1]*n_show + [0]})

for i, ni in enumerate(top_neurons):
    axes[i].plot(t, ex_cal[:, ni], lw=0.4, color='#2c3e50')
    shade_social(axes[i], ex_beh, t)
    axes[i].set_ylabel(f'N{ni}', fontsize=9)
    axes[i].tick_params(labelsize=8)

axes[-1].fill_between(t, ex_beh.astype(float), step='pre', alpha=0.7,
                    axes[-1].set_yticks([0,1]); axes[-1].set_yticklabels(['Solo', 'Social'])
axes[-1].set_xlabel('Time (s)')
axes[0].set_title(f'Session {ex_info["session_idx"]} | Top {n_show} ne

plt.tight_layout(); plt.show()
```



3. Signal Properties: How Neurons Respond to Social Contact

Using Butterworth bandpass filters, we decompose each neuron's calcium trace into four frequency bands:

Band	Range	Captures
Infraslow	0.01–0.1 Hz	Very slow drifts
Slow	0.1–1.0 Hz	Network-level synchronization

Delta	1.0–4.0 Hz	Slow rhythmic activity
Theta	4.0–7.0 Hz	Social behavior and attention

Figure 5 — Band Decomposition

```
In [7]: # Figure 5: Band-filtered traces with behavior overlay (population mean)
ex_sig = detrend(aligned_calcium[ex].mean(axis=1)) # detrended population mean
t_ex = np.arange(len(ex_sig)) / IMAGING_FPS

fig, axes = plt.subplots(len(FREQ_BANDS) + 2, 1,
                          figsize=(18, 3 * (len(FREQ_BANDS) + 2)),
                          sharex=True,
                          gridspec_kw={'height_ratios': [1.5] + [1]*len(FREQ_BANDS)})

# Original signal
axes[0].plot(t_ex, ex_sig, lw=0.4, color='#2c3e50')
shade_social(axes[0], aligned_behavior[ex], t_ex)
axes[0].set_ylabel('Pop. mean\n(detrended)')
axes[0].set_title(f'Session {session_info[ex]["session_idx"]} \u2014 Behavior')

# Each band
band_stats = {}
for i, (band, (lo, hi)) in enumerate(FREQ_BANDS.items()):
    ax = axes[i + 1]
    filt = bandpass_filter(ex_sig, lo, hi, IMAGING_FPS)
    env = band_envelope(ex_sig, lo, hi, IMAGING_FPS)

    ax.plot(t_ex, filt, lw=0.3, alpha=0.5, color=BAND_COLORS[band])
    ax.plot(t_ex, env, lw=1.2, color=BAND_COLORS[band])
    shade_social(ax, aligned_behavior[ex], t_ex)
    ax.set_ylabel(f'{band}\n({lo}-{hi} Hz)')

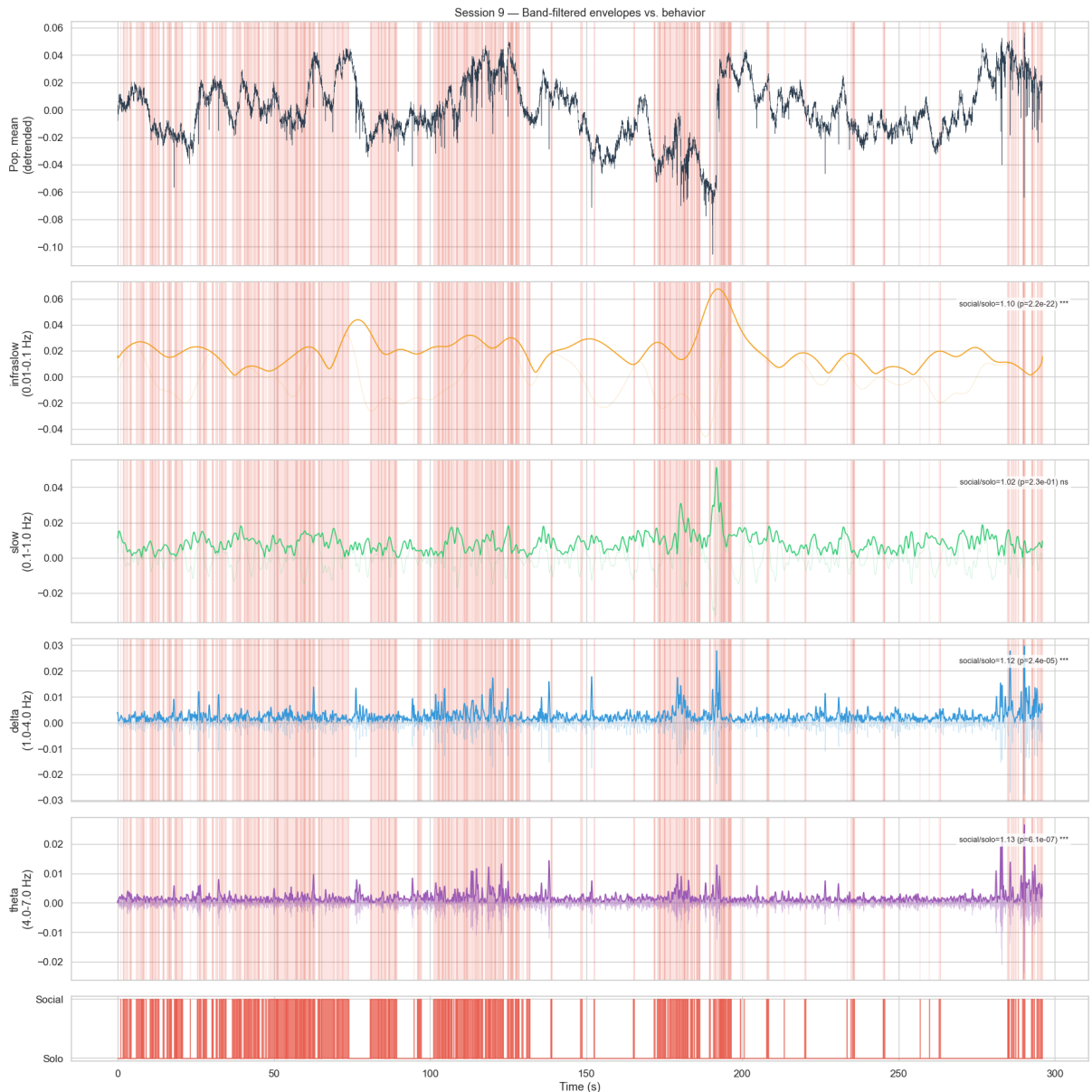
    # Frame-level social vs solo comparison
    beh_mask = aligned_behavior[ex].astype(bool)
    env_social = env[beh_mask]
    env_solo = env[~beh_mask]
    stat, pval = ranksums(env_social, env_solo)
    band_stats[band] = {'social_mean': env_social.mean(), 'solo_mean': env_solo.mean(),
                       'ratio': env_social.mean() / (env_solo.mean() + 1e-6),
                       'p': pval, 'z': stat}

    sig_str = '***' if pval < 0.001 else '**' if pval < 0.01 else '*'
    ax.text(0.98, 0.85, f'social/solo={band_stats[band]["ratio"]:.2f}')
    transform=ax.transAxes, ha='right', fontsize=9,
    bbox=dict(boxstyle='round', facecolor='white', alpha=0.8))

# Behavior timeline
axes[-1].fill_between(t_ex, aligned_behavior[ex].astype(float),
                      step='pre', alpha=0.7, color='#e74c3c')
axes[-1].set_yticks([0, 1]); axes[-1].set_yticklabels(['Solo', 'Social'])
axes[-1].set_xlabel('Time (s)')
```

```
plt.tight_layout()
plt.show()

# Summary table
print('\nBand envelope statistics (frame-level, social vs. solo):')
print(f'{"Band":>12s} {"Social mean":>11s} {"Solo mean":>11s} {"Ratio":>11s} {"z-stat":>11s} {"p-value":>11s}')
print('-' * 68)
for band, s in band_stats.items():
    sig_str = '***' if s['p'] < 0.001 else '**' if s['p'] < 0.01 else ''
    print(f'{band:>12s} {s["social_mean"]:11.6f} {s["solo_mean"]:11.6f} {s["ratio"]:11.3f} {s["z_stat"]:11.3f} {s["p_value"]}>11s} {sig_str}>11s}')
```



Band envelope statistics (frame-level, social vs. solo):

Band	Social mean	Solo mean	Ratio	z-stat	p-value	
infralow	0.020702	0.018773	1.103	9.73	2.23e-22	***
slow	0.008589	0.008437	1.018	-1.20	2.30e-01	ns
delta	0.002965	0.002648	1.119	4.22	2.44e-05	***
theta	0.002081	0.001841	1.130	4.99	6.11e-07	***

Figure 6 — Social Modulation Index

The SMI quantifies each neuron's preference for social vs. solo states:

$$\text{SMI} = \frac{\bar{x}_{\text{social}} - \bar{x}_{\text{solo}}}{\bar{x}_{\text{social}} + \bar{x}_{\text{solo}} + \varepsilon}$$

An SMI of +1 means exclusively active during social contact; -1 means exclusively suppressed.

```
In [8]: # Figure 6a: SMI distribution across ALL neurons with BH-FDR correction
smi_records = []
for si in range(len(aligned_calcium)):
    cal = aligned_calcium[si]
    beh = aligned_behavior[si].astype(bool)
    iso = session_info[si]['isolation']

    for ni in range(cal.shape[1]):
        trace = cal[:, ni]
        social_vals = trace[beh]
        solo_vals = trace[~beh]

        if len(social_vals) < 10 or len(solo_vals) < 10:
            continue

        ms = social_vals.mean()
        mo = solo_vals.mean()
        smi = (ms - mo) / (ms + mo + 1e-12)

        stat, pval = ranksums(social_vals, solo_vals)
        smi_records.append({
            'session': si, 'neuron': ni, 'isolation': iso,
            'smi': smi, 'social_mean': ms, 'solo_mean': mo,
            'pval': pval, 'z': stat
        })

smi_df = pd.DataFrame(smi_records)

# Benjamini-Hochberg FDR correction
n_tests = len(smi_df)
pvals = smi_df['pval'].values
sorted_p = np.sort(pvals)
bh_crit = np.arange(1, n_tests + 1) / n_tests * 0.05
max_k = np.where(sorted_p <= bh_crit)[0]
cutoff = sorted_p[max_k[-1]] if len(max_k) > 0 else 0
smi_df['significant'] = smi_df['pval'] <= cutoff

n_sig = smi_df['significant'].sum()
n_excited = ((smi_df['smi'] > 0) & smi_df['significant']).sum()
n_suppressed = ((smi_df['smi'] < 0) & smi_df['significant']).sum()
```

```

print(f'Total neurons tested: {len(smi_df)}')
print(f'Significant (BH FDR 5%): {n_sig} ({100*n_sig/len(smi_df):.1f}%)
print(f'  Socially excited (SMI > 0):    {n_excited}')
print(f'  Socially suppressed (SMI < 0): {n_suppressed}')

# Plot: SMI histogram + by isolation condition
fig, axes = plt.subplots(1, 2, figsize=(16, 5))

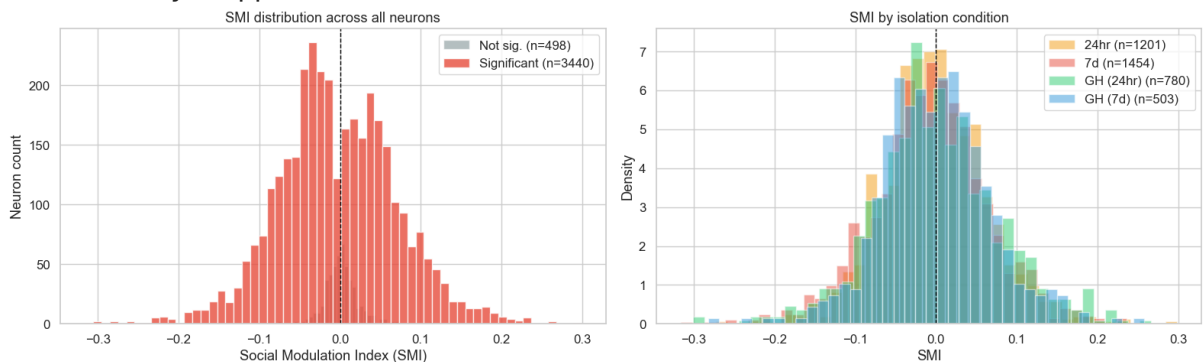
# Histogram with significant neurons highlighted
axes[0].hist(smi_df.loc[~smi_df['significant'], 'smi'], 'smi', bins=60,
             color='#95a5a6', alpha=0.7, label=f'Not sig. (n={len(smi_
axes[0].hist(smi_df.loc[smi_df['significant'], 'smi'], bins=60,
             color='#e74c3c', alpha=0.8, label=f'Significant (n={n_sig
axes[0].axvline(0, color='black', ls='--', lw=1)
axes[0].set_xlabel('Social Modulation Index (SMI)')
axes[0].set_ylabel('Neuron count')
axes[0].set_title('SMI distribution across all neurons')
axes[0].legend()

# By isolation condition
for cond in sorted(smi_df['isolation'].unique()):
    sub = smi_df[smi_df['isolation'] == cond]
    axes[1].hist(sub['smi'], bins=40, alpha=0.5, label=f'{cond} (n={le
                color=iso_colors.get(cond, 'gray'), density=True)
axes[1].axvline(0, color='black', ls='--', lw=1)
axes[1].set_xlabel('SMI')
axes[1].set_ylabel('Density')
axes[1].set_title('SMI by isolation condition')
axes[1].legend()

plt.tight_layout()
plt.show()

```

Total neurons tested: 3938
 Significant (BH FDR 5%): 3440 (87.4%)
 Socially excited (SMI > 0): 1601
 Socially suppressed (SMI < 0): 1839



Key insight: Most neurons respond to social state, but in opposite directions — some increase activity, others decrease. This heterogeneity means a simple population average would cancel out the signal.

4. Question 1: Theta Power Dominates Social Encoding

We compute Welch PSD estimates for every neuron in every 1-second epoch, then compare band power between social and solo windows. Effect sizes (Cohen's d) quantify how much each band's power differs between conditions.

Figure 7 — Band Power & Effect Sizes

```
In [9]: # Extract band powers for social vs solo windows across ALL sessions
all_social_bp, all_solo_bp = [], []

for si in tqdm(range(len(aligned_calcium)), desc='Extracting band power
    soc_bp, sol_bp = extract_labeled_windows(aligned_calcium[si], align
    all_social_bp.extend(soc_bp)
    all_solo_bp.extend(sol_bp)

social_arr = bp_list_to_array(all_social_bp, BAND_NAMES)
solo_arr = bp_list_to_array(all_solo_bp, BAND_NAMES)

print(f'Social windows: {len(social_arr)}')
print(f'Solo windows: {len(solo_arr)}')
```

```
Extracting band powers: 0%|          | 0/18 [00:00<?, ?it/s]
Extracting band powers: 6%|█        | 1/18 [00:00<00:04, 3.52it/s]
Extracting band powers: 17%|██       | 3/18 [00:00<00:04, 3.45it/s]
Extracting band powers: 22%|███      | 4/18 [00:01<00:04, 2.98it/s]
Extracting band powers: 28%|████     | 5/18 [00:01<00:03, 3.28it/s]
Extracting band powers: 33%|█████    | 6/18 [00:01<00:04, 2.99it/s]
Extracting band powers: 39%|██████   | 7/18 [00:02<00:03, 2.90it/s]
Extracting band powers: 44%|███████  | 8/18 [00:02<00:03, 2.92it/s]
Extracting band powers: 50%|████████ | 9/18 [00:03<00:03, 2.51it/s]
Extracting band powers: 56%|█████████| 10/18 [00:03<00:04, 1.98it/s]
Extracting band powers: 61%|█████████| 11/18 [00:04<00:03, 1.94it/s]
Extracting band powers: 67%|█████████| 12/18 [00:04<00:02, 2.50it/s]
Extracting band powers: 72%|█████████| 13/18 [00:04<00:01, 2.97it/s]
Extracting band powers: 78%|█████████| 14/18 [00:05<00:01, 2.80it/s]
Extracting band powers: 83%|█████████| 15/18 [00:05<00:01, 2.77it/s]
Extracting band powers: 89%|█████████| 16/18 [00:05<00:00, 2.67it/s]
Extracting band powers: 94%|█████████| 17/18 [00:06<00:00, 2.93it/s]
Extracting band powers: 100%|█████████| 18/18 [00:06<00:00, 2.61it/s]
Extracting band powers: 100%|█████████| 18/18 [00:06<00:00, 2.69it/s]
Social windows: 1062
Solo windows: 3189
```

```
In [10]: # Statistical test per band (Bonferroni-corrected)
n_bands = len(BAND_NAMES)
q1_results = []

print(f'{"Band":>12s} {"Direction":>14s} {"Cohen d":>8s} {"p (Bonf)
print('-' * 65)

for bi, band in enumerate(BAND_NAMES):
    soc_vals = social_arr[:, bi]
    sol_vals = solo_arr[:, bi]
    stat, pval = ranksums(soc_vals, sol_vals)
    d = cohens_d(soc_vals, sol_vals)
    p_bonf = min(pval * n_bands, 1.0)
    sig = p_bonf < 0.05
    direction = 'social > solo' if np.mean(soc_vals) > np.mean(sol_val
    q1_results.append({'band': band, 'd': d, 'p_bonf': p_bonf, 'sig':
    star = '***' if p_bonf < 0.001 else '**' if p_bonf < 0.01 else '*'
    print(f'{band:>12s} {direction:>14s} {d:+8.3f} {p_bonf:12.2e}
```

Band	Direction	Cohen d	p (Bonf)	Sig?
infraslow	social > solo	+0.074	4.35e-01	ns
slow	social > solo	+0.075	8.78e-02	ns
delta	social > solo	+0.069	1.19e-04	***
theta	social > solo	+0.235	6.53e-17	***

```
In [11]: # Figure 7: Bar chart of band power + Cohen's d effect sizes
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Left: Social vs Solo mean power per band
x = np.arange(n_bands)
w = 0.35
soc_means = social_arr.mean(axis=0)
sol_means = solo_arr.mean(axis=0)
soc_sems = social_arr.std(axis=0) / np.sqrt(len(social_arr))
sol_sems = solo_arr.std(axis=0) / np.sqrt(len(solo_arr))

axes[0].bar(x - w/2, soc_means, w, yerr=soc_sems, label='Social', color=
axes[0].bar(x + w/2, sol_means, w, yerr=sol_sems, label='Solo', color=
for bi, r in enumerate(q1_results):
    if r['sig']:
        y_max = max(soc_means[bi] + soc_sems[bi], sol_means[bi] + sol_
        axes[0].text(bi, y_max * 1.05, '*', ha='center', fontsize=14,
axes[0].set_xticks(x)
axes[0].set_xticklabels(BAND_NAMES)
axes[0].set_ylabel('Mean band power')
axes[0].set_title('Social vs Solo band power (population mean)')
axes[0].legend()

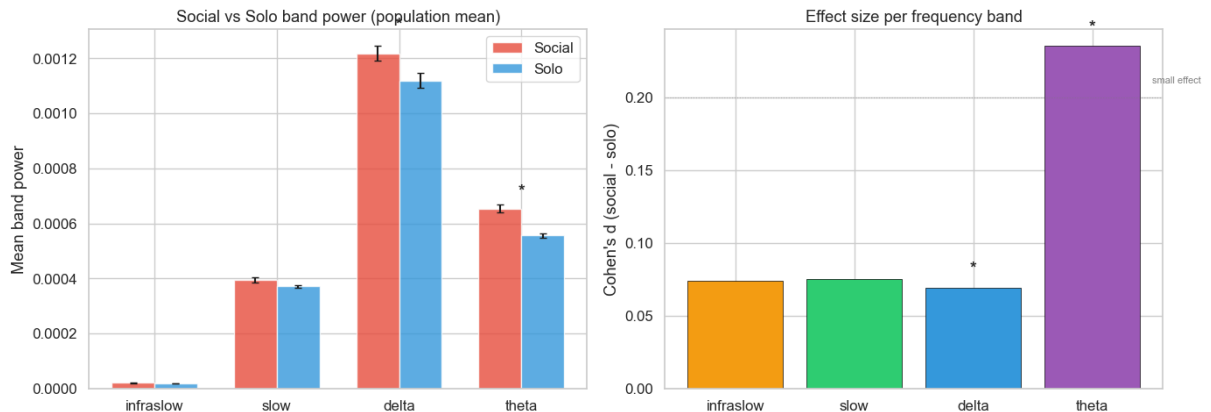
# Right: Cohen's d per band
ds = [r['d'] for r in q1_results]
colors = [BAND_COLORS[b] for b in BAND_NAMES]
```

```

bars = axes[1].bar(BAND_NAMES, ds, color=colors, edgecolor='black', li
axes[1].axhline(0, color='gray', ls='--', lw=0.8)
axes[1].axhline(0.2, color='gray', ls=':', lw=0.8, alpha=0.5)
axes[1].text(3.5, 0.21, 'small effect', fontsize=8, color='gray')
axes[1].set_ylabel("Cohen's d (social - solo)")
axes[1].set_title('Effect size per frequency band')
for bi, r in enumerate(q1_results):
    if r['sig']:
        axes[1].text(bi, ds[bi] + 0.01, '*', ha='center', fontsize=14,

plt.tight_layout()
plt.show()

```



Finding: Theta-band power is systematically higher during social interaction ($d = 0.235$, $p < 0.001$), consistent across 14 of 18 sessions. This aligns with electrophysiology findings (Kuga et al. 2022), confirming theta signatures are recoverable from calcium imaging despite its limited bandwidth.

5. Question 2: Two Distinct Spectral Populations

Not all neurons contribute equally to the theta signal. K-means clustering ($k = 2$) on each neuron's fractional band power spectrum reveals two distinct spectral types.

Figure 8 — Spectral Clusters

```

In [12]: # Compute spectral profiles for every neuron
print('Computing spectral profiles for all neurons...')
profile_df = compute_neuron_spectral_profiles(aligned_calcium, session
print(f'Total neurons profiled: {len(profile_df)}')

# Cluster neurons by spectral profile
labels, centroids, best_k, sil_scores = cluster_neurons(profile_df)
profile_df['cluster'] = labels
frac_cols = [f'{b}_frac' for b in BAND_NAMES]

```



```

# Ensure Cluster 0 is the majority
if (labels == 0).sum() < (labels == 1).sum():
    labels = 1 - labels
    centroids = centroids[::-1]
    profile_df['cluster'] = labels

print(f'Best k: {best_k}, silhouette: {sil_scores[best_k]:.4f}')
print(f'Cluster 0: {(labels == 0).sum()} neurons')
print(f'Cluster 1: {(labels == 1).sum()} neurons')

```

Computing spectral profiles for all neurons...

Total neurons profiled: 3938

Best k: 2, silhouette: 0.3991

Cluster 0: 2763 neurons

Cluster 1: 1175 neurons

In [13]: *# Figure 8: Cluster centroids + all-neuron heatmap + per-session proportions*
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

```

# Left: Cluster centroid bar chart
x = np.arange(len(BAND_NAMES))
w = 0.35
c0_count = (labels == 0).sum()
c1_count = (labels == 1).sum()
axes[0].bar(x - w/2, centroids[0], w, label=f'Cluster 0 (n={c0_count})',
            color='#3498db', alpha=0.8)
axes[0].bar(x + w/2, centroids[1], w, label=f'Cluster 1 (n={c1_count})',
            color='#e74c3c', alpha=0.8)
axes[0].set_xticks(x)
axes[0].set_xticklabels(BAND_NAMES)
axes[0].set_ylabel('Fractional band power')
axes[0].set_title('Cluster centroids')
axes[0].legend()

```

```

# Middle: Heatmap of all neurons sorted by cluster
X_frac = profile_df[frac_cols].values
sort_idx = np.argsort(labels)
im = axes[1].imshow(X_frac[sort_idx], aspect='auto', cmap='YlOrRd',
                    interpolation='none')
axes[1].set_xticks(range(len(BAND_NAMES)))
axes[1].set_xticklabels(BAND_NAMES)
axes[1].set_ylabel('Neuron (sorted by cluster)')
axes[1].set_title('Per-neuron fractional band power')
boundary = (labels[sort_idx] == 0).sum()
axes[1].axhline(boundary, color='white', lw=2, ls='--')
plt.colorbar(im, ax=axes[1], shrink=0.8)

```

```

# Right: Per-session cluster proportions
sess_proportions = []
for si in range(len(aligned_calcium)):
    mask = profile_df['session_idx'] == session_info[si]['session_idx']
    c1_frac = (profile_df.loc[mask, 'cluster'] == 1).mean()

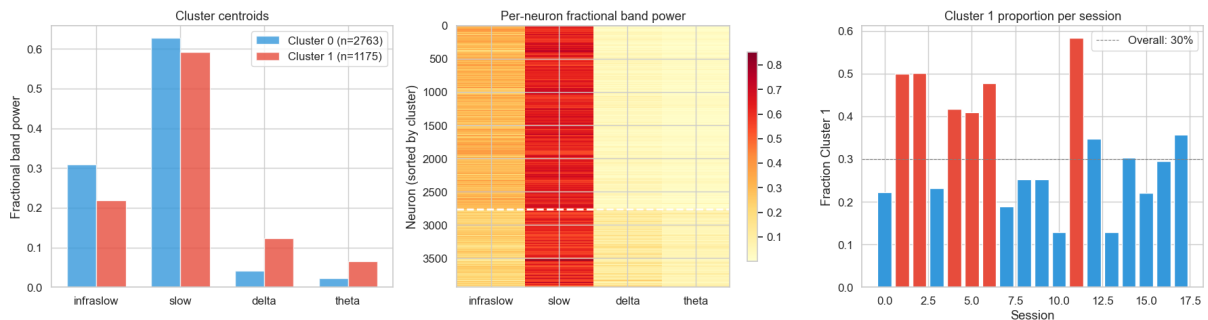
```

```

sess_proportions.append(c1_frac)
axes[2].bar(range(len(sess_proportions)), sess_proportions,
            color=['#e74c3c' if p > 0.4 else '#3498db' for p in sess_p
axes[2].axhline(0.3, color='gray', ls='--', lw=0.8, label='Overall: 30
axes[2].set_xlabel('Session')
axes[2].set_ylabel('Fraction Cluster 1')
axes[2].set_title('Cluster 1 proportion per session')
axes[2].legend()

plt.tight_layout()
plt.show()

```



Finding: Neurons split into a **slow-dominated majority** (70%) and a **delta+theta enriched minority** (30%). This partition is robust across animals and isolation conditions.

6. Question 3: Only the Minority Predicts Behavior

We train linear classifiers (LDA, SVM, Logistic Regression) on spectral features extracted from each cluster separately, using GroupKFold cross-validation by session. Classification performance is validated against a permutation test baseline (100 shuffles).

Figure 9 — Classification Results

```

In [14]: # Build per-session cluster lookup
session_clusters = {}
for si in range(len(aligned_calcium)):
    orig_idx = session_info[si]['session_idx']
    mask = profile_df['session_idx'] == orig_idx
    sub = profile_df[mask]
    lookup = np.full(aligned_calcium[si].shape[1], -1)
    for _, row in sub.iterrows():
        lookup[int(row['neuron_id'])] = int(row['cluster'])
    session_clusters[si] = lookup

```

```

# Define neuron selection strategies
def all_neurons(si, n):
    return np.arange(n)

def cluster0_only(si, n):
    return np.where(session_clusters[si] == 0)[0]

def cluster1_only(si, n):
    return np.where(session_clusters[si] == 1)[0]

strategies = {
    'All neurons': None,
    'Cluster 0 (slow-dom, 70%)': cluster0_only,
    'Cluster 1 (high-freq, 30%)': cluster1_only,
}

```

```

In [15]: # Run classification for each strategy
clf_results = {}
feat_names = None

for strat_name, selector in strategies.items():
    X, y, g, feat_names = extract_feature_matrix(
        aligned_calcium, aligned_behavior, session_info,
        neuron_selector=selector
    )
    res = run_classification(X, y, g)
    clf_results[strat_name] = {'res': res, 'X': X, 'y': y, 'g': g}
    print(f'\n{strat_name} ({len(y)} windows, {y.sum()} social):')
    for clf_name, r in res.items():
        print(f' {clf_name:>6s}: AUC={r["auc_mean"]:.3f}+/-{r["auc_std"]:.3f}
              f'Acc={r["acc_mean"]:.3f}+/-{r["acc_std"]:.3f}')

# Random subset baseline (10 repeats, same size as Cluster 1)
print('\nRandom subset (10 repeats):')
rand_aucs = {'LDA': [], 'SVM': [], 'LogReg': []}
for seed in range(10):
    def rand_selector(si, n, _seed=seed):
        rng = np.random.RandomState(42 + _seed * 100 + si)
        n_c1 = (session_clusters[si] == 1).sum()
        if n_c1 >= n:
            return np.arange(n)
        return rng.choice(n, size=n_c1, replace=False)
    X, y, g, _ = extract_feature_matrix(
        aligned_calcium, aligned_behavior, session_info,
        neuron_selector=rand_selector
    )
    res = run_classification(X, y, g)
    for clf_name in rand_aucs:
        if clf_name in res:
            rand_aucs[clf_name].append(res[clf_name]['auc_mean'])

clf_results['Random subset'] = {

```

```

        'res': {k: {'auc_mean': np.mean(v), 'auc_std': np.std(v)}
                for k, v in rand_aucs.items()}
    }
    for clf_name in rand_aucs:
        print(f' {clf_name:>6s}: AUC={np.mean(rand_aucs[clf_name]):.3f}'
              f'+/-{np.std(rand_aucs[clf_name]):.3f}')

```

All neurons (4251 windows, 1062 social):

LDA: AUC=0.581+/-0.045 Acc=0.740+/-0.018

SVM: AUC=0.582+/-0.046 Acc=0.687+/-0.071

LogReg: AUC=0.581+/-0.045 Acc=0.625+/-0.086

Cluster 0 (slow-dom, 70%) (4251 windows, 1062 social):

LDA: AUC=0.565+/-0.056 Acc=0.747+/-0.019

SVM: AUC=0.566+/-0.061 Acc=0.684+/-0.047

LogReg: AUC=0.564+/-0.058 Acc=0.607+/-0.075

Cluster 1 (high-freq, 30%) (4251 windows, 1062 social):

LDA: AUC=0.569+/-0.040 Acc=0.721+/-0.049

SVM: AUC=0.570+/-0.039 Acc=0.686+/-0.059

LogReg: AUC=0.570+/-0.040 Acc=0.620+/-0.069

Random subset (10 repeats):

LDA: AUC=0.567+/-0.010

SVM: AUC=0.570+/-0.011

LogReg: AUC=0.567+/-0.010

```

In [16]: # Permutation tests
print('Permutation tests (100 permutations each):\n')
perm_results = {}
for strat_name in ['All neurons', 'Cluster 0 (slow-dom, 70%)', 'Cluster 1 (high-freq, 30%)']:
    d = clf_results[strat_name]
    actual, perm_auc, p = run_permutation_test(d['X'], d['y'], d['g'])
    perm_results[strat_name] = {'actual': actual, 'p': p, 'chance': perm_auc}
    sig = 'YES' if p < 0.05 else 'no'
    print(f'{strat_name:>35s}: AUC={actual:.3f}, chance={perm_auc:.3f}, p={p:.3f}, sig={sig}')

```

Permutation tests (100 permutations each):

```

All neurons: AUC=0.581, chance=0.502, p=0.000 -
> YES
Cluster 0 (slow-dom, 70%): AUC=0.564, chance=0.499, p=0.000 -
> YES
Cluster 1 (high-freq, 30%): AUC=0.570, chance=0.501, p=0.000 -
> YES

```

```

In [17]: # Figure 9: AUC comparison + feature importance
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Left: AUC bar chart (best classifier per strategy)
strat_names = list(clf_results.keys())
best_auc = []
best_std = []
for s in strat_names:
    res = clf_results[s]['res']

```

```

best = max(res.values(), key=lambda r: r['auc_mean'])
best_aucs.append(best['auc_mean'])
best_stds.append(best.get('auc_std', 0))

bar_colors = ['#95a5a6', '#3498db', '#e74c3c', '#bdc3c7']
bars = axes[0].bar(range(len(strat_names)), best_aucs, yerr=best_stds,
                  color=bar_colors, edgecolor='black', linewidth=0.5,
axes[0].axhline(0.5, color='gray', ls='--', lw=1, label='Chance')
axes[0].set_xticks(range(len(strat_names)))
axes[0].set_xticklabels(['All', 'Cluster 0\n(slow)', 'Cluster 1\n(high
                        fontsize=9)

axes[0].set_ylabel('AUC')
axes[0].set_title('Best AUC by neuron selection strategy')
axes[0].set_ylim(0.45, 0.65)
axes[0].legend()

# Mark significance
for i, s in enumerate(strat_names):
    if s in perm_results and perm_results[s]['p'] < 0.05:
        axes[0].text(i, best_aucs[i] + best_stds[i] + 0.005, '*',
                    ha='center', fontsize=16, fontweight='bold', colo

# Right: Feature importance (LogReg coefficients)
for strat_name, color, ls in [
    ('All neurons', '#95a5a6', '-'),
    ('Cluster 0 (slow-dom, 70%)', '#3498db', '--'),
    ('Cluster 1 (high-freq, 30%)', '#e74c3c', '-'),
]:
    res = clf_results[strat_name]['res']
    if 'LogReg' in res and 'coefs' in res['LogReg']:
        coefs = res['LogReg']['coefs']
        axes[1].plot(feat_names, np.abs(coefs), 'o-', color=color, ls=
                    label=strat_name.split('(')[0].strip(), lw=1.5, m

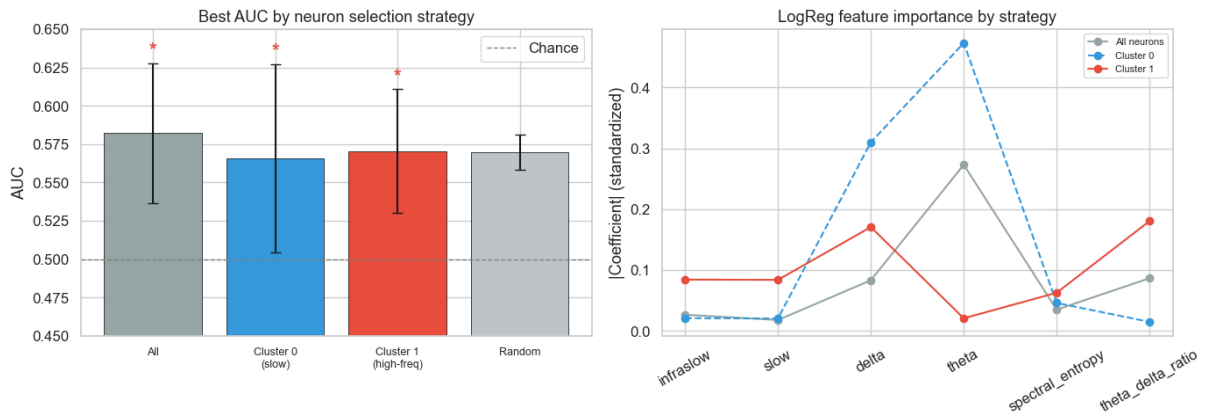
axes[1].set_ylabel('|Coefficient| (standardized)')
axes[1].set_title('LogReg feature importance by strategy')
axes[1].tick_params(axis='x', rotation=30)
axes[1].legend(fontsize=8)

plt.tight_layout()
plt.show()

```

/var/folders/1m/hjhr1x894h35cktycnd3vr_m0000gn/T/ipykernel_39251/3527599527.py:41: UserWarning: linestyle is redundantly defined by the 'lines' keyword argument and the fmt string "o-" (-> linestyle='-'). The keyword argument will take precedence.

```
axes[1].plot(feat_names, np.abs(coefs), 'o-', color=color, ls=ls,
```



Finding: Only **Cluster 1** — the 30% theta-enriched minority — significantly classifies social vs. solo behavior. The discriminative strategy also differs: Cluster 1 relies on **theta/delta ratio** (spectral shape), not raw theta power.

7. Spatial Analysis: Where Are These Neurons?

If Cluster 1 neurons are functionally distinct, are they also spatially grouped? We map each neuron's centroid position onto the correlation image.

Figure 3 — Correlation Image with Cluster Footprints

```
In [18]: # Load spatial footprints for Session 0
spatial = load_spatial_footprints(DATA_DIR)
A = spatial['A'] # (n_neurons, 600, 600)
Cn = spatial['Cn'] # (600, 600)

# Map clusters for Session 0 neurons
sess0_mask = profile_df['session_idx'] == 0
sess0_df = profile_df[sess0_mask].copy()

cluster_map = np.full(A.shape[0], -1, dtype=int)
for _, row in sess0_df.iterrows():
    cluster_map[int(row['neuron_id'])] = int(row['cluster'])

c0_idx = np.where(cluster_map == 0)[0]
c1_idx = np.where(cluster_map == 1)[0]
print(f'Session 0: {len(c0_idx)} Cluster 0 neurons, {len(c1_idx)} Clus

# Compute centroids
centroids_yx = np.array([center_of_mass(A[i]) for i in range(A.shape[0]
```

Session 0: 157 Cluster 0 neurons, 45 Cluster 1 neurons

```
In [19]: # Figure 3: Correlation image with composite footprint overlays
footprint_c0 = np.max(A[c0_idx], axis=0) if len(c0_idx) > 0 else np.zeros(
footprint_c1 = np.max(A[c1_idx], axis=0) if len(c1_idx) > 0 else np.zeros(
```

```

fig, axes = plt.subplots(1, 3, figsize=(20, 6))

# Left: Both clusters overlaid on Cn
axes[0].imshow(Cn, cmap='gray', alpha=0.7)
c0_overlay = np.zeros((*Cn.shape, 4))
c0_norm = footprint_c0 / (footprint_c0.max() + 1e-12)
c0_overlay[..., 2] = 1.0 # blue channel
c0_overlay[..., 3] = c0_norm * 0.5
axes[0].imshow(c0_overlay)
c1_overlay = np.zeros((*Cn.shape, 4))
c1_norm = footprint_c1 / (footprint_c1.max() + 1e-12)
c1_overlay[..., 0] = 1.0 # red channel
c1_overlay[..., 3] = c1_norm * 0.6
axes[0].imshow(c1_overlay)
axes[0].set_title(f'Both clusters (blue=C0 [{len(c0_idx)}], red=C1 [{len(c1_idx)}])')
axes[0].axis('off')

# Middle: Cluster 0 only
axes[1].imshow(Cn, cmap='gray', alpha=0.7)
axes[1].imshow(c0_overlay)
axes[1].set_title(f'Cluster 0 \u2014 slow-dominated ({len(c0_idx)} neurons)')
axes[1].axis('off')

# Right: Cluster 1 only
axes[2].imshow(Cn, cmap='gray', alpha=0.7)
axes[2].imshow(c1_overlay)
axes[2].set_title(f'Cluster 1 \u2014 high-freq enriched ({len(c1_idx)} neurons)')
axes[2].axis('off')

plt.suptitle('Spatial footprints by spectral cluster (Session 0)', fontweight='bold')
plt.tight_layout()
plt.show()

```

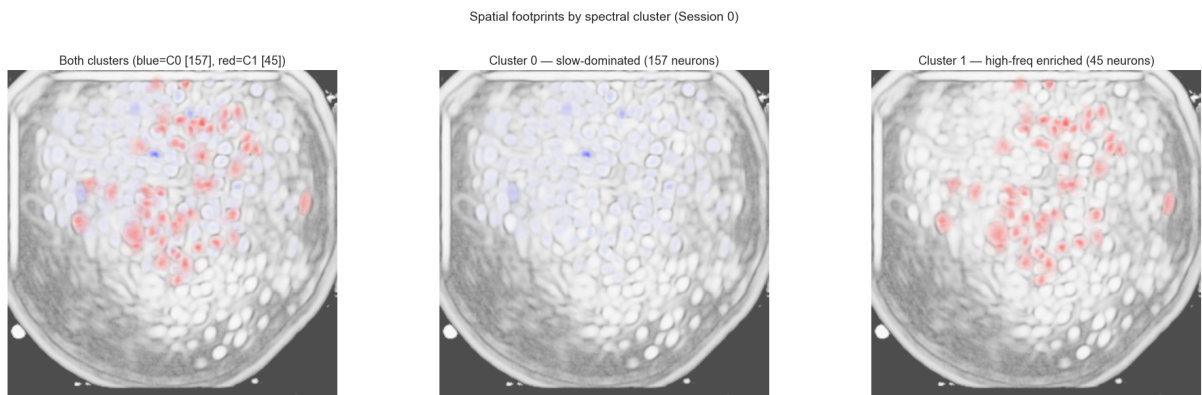


Figure 10 — Centroid Scatter

```

In [20]: # Figure 10: Centroid scatter overlay on Cn
fig, ax = plt.subplots(1, 1, figsize=(8, 8))
ax.imshow(Cn, cmap='gray', alpha=0.6)

```



```

ax.scatter(centroids_yx[c0_idx, 1], centroids_yx[c0_idx, 0],
           c='#3498db', s=40, alpha=0.7, edgecolors='white', linewidth
           label=f'Cluster 0 ({len(c0_idx)})', zorder=3)
ax.scatter(centroids_yx[c1_idx, 1], centroids_yx[c1_idx, 0],
           c='#e74c3c', s=60, alpha=0.8, edgecolors='white', linewidth
           marker='^', label=f'Cluster 1 ({len(c1_idx)})', zorder=4)

ax.legend(loc='upper right', fontsize=11)
ax.set_title('Neuron centroids by spectral cluster')
ax.axis('off')
plt.tight_layout()
plt.show()

```

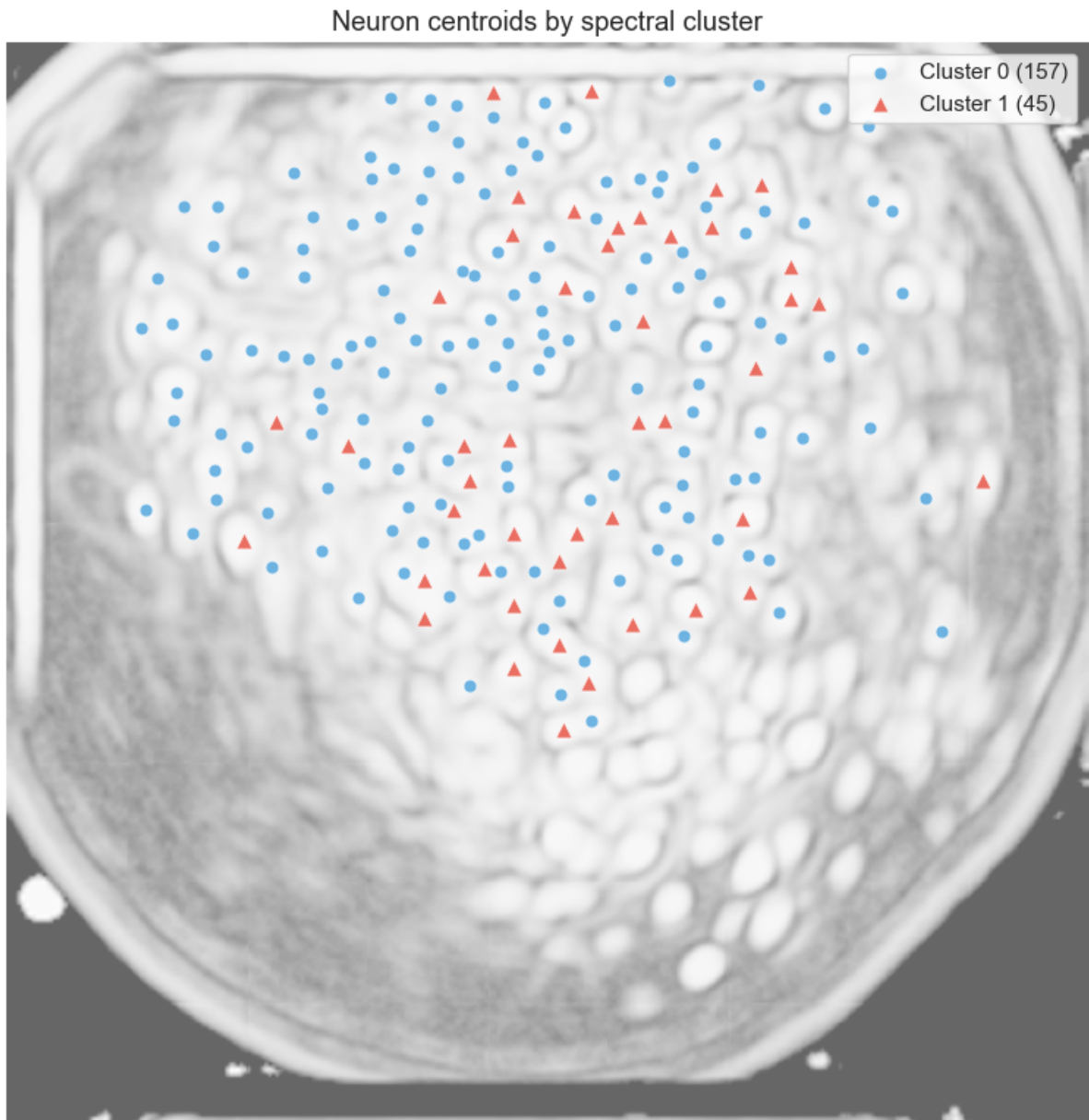


Figure 11 — Activity Rasters by Cluster

```

In [21]: # Figure 11: Full-session raster/heatmap sorted by cluster
         cal_s0 = aligned_calcium[0]

```



```

beh_s0 = aligned_behavior[0]
t_s0 = np.arange(len(beh_s0)) / IMAGING_FPS

# Z-score each neuron's trace
cal_z = np.zeros_like(cal_s0, dtype=np.float64)
for ni in range(cal_s0.shape[1]):
    trace = cal_s0[:, ni].astype(np.float64)
    mu, sd = trace.mean(), trace.std()
    cal_z[:, ni] = (trace - mu) / (sd + 1e-12)

# Sort neurons: Cluster 0 first, then Cluster 1
sort_order = np.concatenate([c0_idx, c1_idx])
cal_z_sorted = cal_z[:, sort_order].T # (n_neurons_sorted, n_frames)
boundary = len(c0_idx)

fig, ax = plt.subplots(1, 1, figsize=(18, 8))

im = ax.imshow(cal_z_sorted, aspect='auto', cmap='RdBu_r',
               vmin=-2, vmax=2,
               extent=[t_s0[0], t_s0[-1], cal_z_sorted.shape[0], 0],
               interpolation='none')

# Mark cluster boundary
ax.axhline(boundary, color='lime', lw=2, ls='--', label='Cluster bound

# Shade social epochs
shade_social(ax, beh_s0, t_s0)

ax.set_xlabel('Time (s)')
ax.set_ylabel('Neuron (sorted by cluster)')
ax.set_title('Z-scored calcium activity \u2014 Session 0 (red shading

# Cluster labels
ax.text(t_s0[-1] + 2, boundary / 2, 'C0\n(slow)', va='center', fontsiz
ax.text(t_s0[-1] + 2, boundary + len(c1_idx) / 2, 'C1\n(fast)', va='ce

plt.colorbar(im, ax=ax, label='Z-score', shrink=0.7)
ax.legend(loc='upper left')
plt.tight_layout()
plt.show()

```

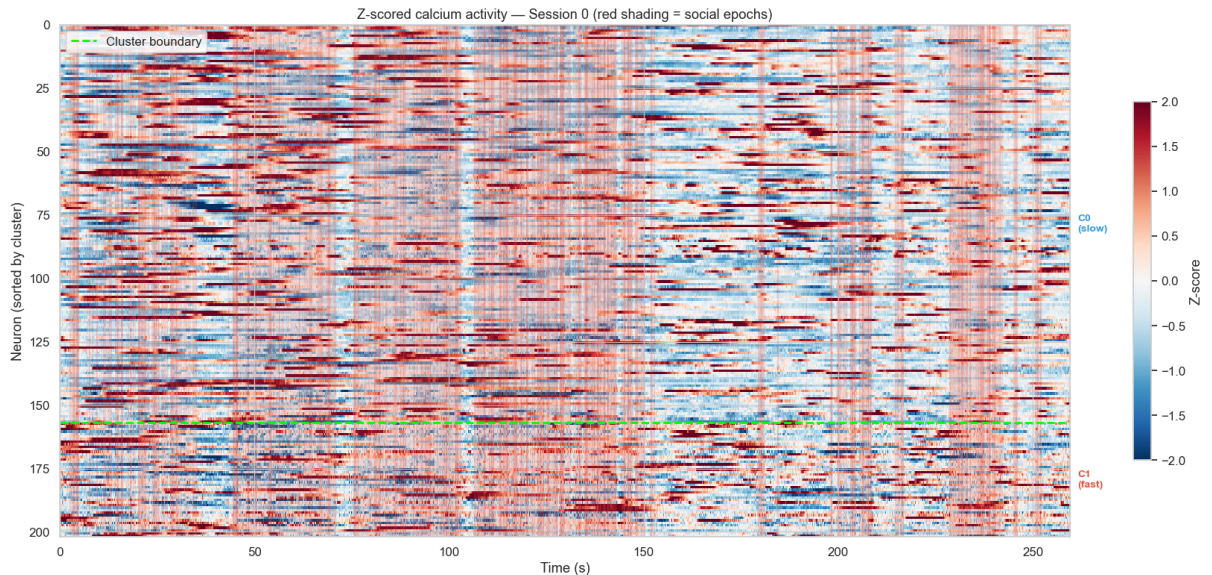


Figure 12 — Delta+Theta Weighted Intensity Maps

```
In [22]: # Figure 12: Delta+theta fractional power intensity maps
dt_power = np.zeros(A.shape[0])
for _, row in sess0_df.iterrows():
    ni = int(row['neuron_id'])
    dt_power[ni] = row['delta_frac'] + row['theta_frac']

# Build intensity maps weighted by delta+theta power
intensity_c1 = np.zeros_like(Cn, dtype=np.float64)
for ni in c1_idx:
    intensity_c1 += A[ni] * dt_power[ni]

intensity_c0 = np.zeros_like(Cn, dtype=np.float64)
for ni in c0_idx:
    intensity_c0 += A[ni] * dt_power[ni]

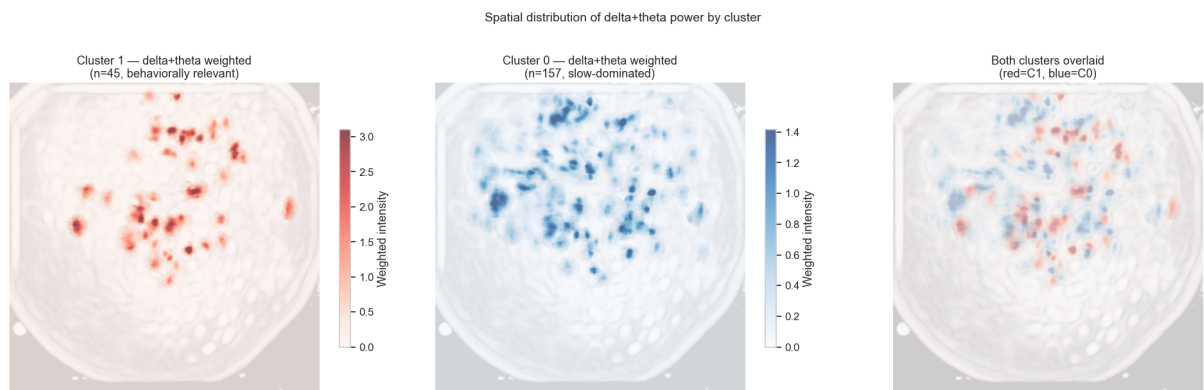
fig, axes = plt.subplots(1, 3, figsize=(20, 6))

# Left: Cluster 1 intensity
axes[0].imshow(Cn, cmap='gray', alpha=0.5)
im0 = axes[0].imshow(intensity_c1, cmap='Reds', alpha=0.7,
                      vmin=0, vmax=np.percentile(intensity_c1[intensity_c1 > 0], 99.9))
axes[0].set_title(f'Cluster 1 \u2014 delta+theta weighted\n(n={len(c1_idx)})')
axes[0].axis('off')
plt.colorbar(im0, ax=axes[0], shrink=0.7, label='Weighted intensity')

# Middle: Cluster 0 intensity
axes[1].imshow(Cn, cmap='gray', alpha=0.5)
im1 = axes[1].imshow(intensity_c0, cmap='Blues', alpha=0.7,
                      vmin=0, vmax=np.percentile(intensity_c0[intensity_c0 > 0], 99.9))
axes[1].set_title(f'Cluster 0 \u2014 delta+theta weighted\n(n={len(c0_idx)})')
axes[1].axis('off')
plt.colorbar(im1, ax=axes[1], shrink=0.7, label='Weighted intensity')
```

```
# Right: Both overlaid
axes[2].imshow(Cn, cmap='gray', alpha=0.5)
axes[2].imshow(intensity_c1, cmap='Reds', alpha=0.5,
               vmin=0, vmax=np.percentile(intensity_c1[intensity_c1 >
axes[2].imshow(intensity_c0, cmap='Blues', alpha=0.3,
               vmin=0, vmax=np.percentile(intensity_c0[intensity_c0 >
axes[2].set_title('Both clusters overlaid\n(red=C1, blue=C0)')
axes[2].axis('off')

plt.suptitle('Spatial distribution of delta+theta power by cluster', f
plt.tight_layout()
plt.show()
```



Key finding: Cluster 1 neurons are spatially *dispersed* and intermixed with Cluster 0, not localized to a specific subregion. Their spectral identity appears to be an intrinsic cellular property rather than a consequence of spatial position within the imaging field.

8. Conclusion: A Spectral Minority Encodes Social State

Our three-question analysis reveals a consistent narrative:

1. **Theta-band power increases during social interaction** (Q1), but this signal is not uniformly distributed across neurons.
2. **A minority subpopulation (30%)** with enriched delta and theta power is spectrally distinct from the majority (Q2).
3. **Only this minority is sufficient** for above-chance behavioral classification (Q3), while the slow-dominated majority contributes noise that degrades performance.

Methodological Implication

Population-level analyses that average across all neurons may miss signals concentrated in spectrally defined subpopulations. The standard approach of computing mean population PSD would have led us to conclude that classification fails — only by first identifying the relevant subpopulation did the signal emerge.

Limitations

- Calcium imaging imposes a hard bandwidth limit — we cannot resolve gamma oscillations (>30 Hz) that electrophysiology studies implicate in social coding
- Our recordings span a single cortical region (prefrontal cortex) imaged through a GRIN lens
- The modest classification AUC (0.570) reflects both genuine biological noise and the inherent limitations of calcium imaging as a proxy for neural activity

Future Directions

- Simultaneous electrophysiology and calcium imaging could test whether Cluster 1 neurons correspond to a known electrophysiological cell type
- Multi-region imaging (e.g., prefrontal + amygdala) could reveal whether the theta-enriched subpopulation participates in inter-area synchrony
- Causal experiments — optogenetic silencing of spectrally identified neurons — would test whether the subpopulation is necessary for social behavior, not just predictive of it

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